Transportation Research Interdisciplinary Perspectives 35 (2026) 101720

Contents lists available at ScienceDirect

Transportation Research Interdisciplinary Perspectives

journal homepage: www.sciencedirect.com/journal/transportation-research-interdisciplinary-perspectives

A holistic continuous approximation framework for strategic last-mile distribution planning☆

Blessy Kirubakaran [a], Anmol Pahwa [a,*], Surendra Reddy Kancharla [b], S. Travis Waller [b]

[a] Civil Engineering Department, Indian Institute of Technology Madras, Chennai 600036, India
[b] "Friedrich List" Faculty of Transport and Traffic Sciences, Technische Universität Dresden 01069 Dresden, Germany

ARTICLE INFO

Keywords:
Vehicle Routing Problem
Continuous Approximation
Adaptive Large Neighborhood Search
Symbolic Regression
Last-Mile Distribution

ABSTRACT

The conventional approach to modelling last-mile distribution operations involves use of the discrete formulation method which renders a representative mathematical model necessitating use of sophisticated solution techniques. While such an intensive approach is justified when decision-makers require a precise plan to support operational planning, this level of precision is redundant for strategic planning, wherein available information is representative but not necessarily exact. To this end, continuous approximation (CA) method offers a practical alternative with use of continuous density functions that estimate parameters approximately, thus striking a balance between estimation accuracy and computational effort. However, typical CA-based routing frameworks assume a simplified distribution environment that significantly limits the capability of such frameworks to accurately model last-mile operations in real-world distribution environments, such as postal service, e-commerce, q-commerce, waste collection, emergency service etc. Thus, the objective of this work is to develop a holistic CA framework for the Vehicle Routing Problem (VRP) to facilitate strategic last-mile distribution planning within diverse logistic settings. Specifically, this study synthesizes a total of 1080 distribution environments, each with peculiar distribution structure and distinctive customer characteristics. Subsequently, it optimizes last-mile distribution for distribution environment instance using an Adaptive Large Neighborhood Search (ALNS) metaheuristic. Finally, this work develops a robust CA-based functional form through symbolic regression to precisely estimate the total distribution tour length across the various distribution environments. In doing so, this work supports strategic decision-making for a wide scope of logistic operators that may have representative but not necessarily exact information.

1. Introduction

Urban logistics plays a pivotal role in supporting the economic needs of a city. It facilitates seamless movement of goods from point of distribution such as warehouses, fulfilment hubs, and delivery centers, to point of consumption including retail stores, homes, and offices. In 2019, urban logistics handled a total of 21.1b parcels worldwide (Universal Postal Union, 2021), and it continues to manage an ever-growing volume of packages due to rapid urbanization and expanding e-retail sector (Anand et al., 2015). However, urban logistics faces numerous challenges in facilitating efficient flow of goods, especially in last-mile distribution – the final leg of the supply-chain. These challenges pertain to the dynamic and stochastic uncertainties of the urban environment, stemming from varying traffic conditions, diverse customer needs, regulatory constraints and more (Mardešić et al., 2023). As a result, the last-mile contributes more than 40 % to the overall supply-chain costs (Capgemini, 2019). This has spurred rigorous inquiries in the field of operations research to optimize last-mile distribution by addressing the complexities inherent to the urban environment.

In this context, the conventional approach to modelling last-mile operations involves use of the discrete formulation method rendering a representative mathematical model (Braekers et al., 2016). This approach, while detailed, necessitates the use of advanced solution techniques due to inherent model complexities such as resource-related constraints, customer-specific time-windows, multi-echelon distribution structures, as well as dynamic and stochastic nature of the delivery

☆ This article is part of a special issue entitled: 'Transforming Freight' published in Transportation Research Interdisciplinary Perspectives.

* Corresponding author.
E-mail addresses: ce24d009@smail.iitm.ac.in (B. Kirubakaran), anmpahwa@civil.iitm.ac.in (A. Pahwa), surendra_reddy.kancharla@tu-dresden.de (S.R. Kancharla), steven_travis.waller@tu-dresden.de (S.T. Waller).

https://doi.org/10.1016/j.trip.2025.101720
Received 1 May 2025; Received in revised form 25 October 2025; Accepted 1 November 2025
Available online 19 December 2025
2590-1982/© 2025 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

Nomenclature

Exogenous variables (given)

l_x: size of the service region (x-axis)
l_y: size of the service region (y-axis)
ρ_x: abscissa of the distribution facility
ρ_y: abscissa of the distribution facility
q_v: capacity of the vehicle operating from the distribution facility
C: set of customers
x_c: abscissa of customer c
y_c: ordinate of customer c
d_c: demand from customer c

Exogenous variables (derived)

n: number of customers
a: effective size of the service region
δ: customer density
s_c: nearest neighbor distance for customer c
ρ_u: unweighted average distance between distribution facility and customers
ρ_w: demand-weighted average distance between distribution facility and customers
μ_d: first moment of customer demand
σ_d: second moment of customer demand
γ_d: third moment of customer demand
κ_d: fourth moment of customer demand
μ_s: first moment of customer nearest neighbor distance
σ_s: second moment of customer nearest neighbor distance
γ_s: third moment of customer nearest neighbor distance
κ_s: fourth moment of customer nearest neighbor distance

Endogenous variable

l: total distribution tour length

environment, among others (Konstantakopoulos et al., 2020; Schneider and Drexl, 2017; Zhang et al., 2021). Given the non-deterministic polynomial-hard nature of such routing problems, the literature has developed a vast array of solution methodologies, most notably using metaheuristic frameworks. These metaheuristic techniques include *local search methods* such as simulated annealing, tabu search, variable neighborhood search, and adaptive large neighborhood search, among others; as well as *evolutionary computation techniques* such as genetic algorithm and evolutionary algorithm; and *swarm intelligence techniques* such as ant-colony optimization, particle swarm optimization, and many more (Elshaer and Awad, 2020; Prodhon and Prins, 2014; Rios et al., 2021). Such an intensive approach is justified for operational planning, wherein decision-makers have exact information and thus require a precise plan to execute specific last-mile distribution tasks. However, for strategic planning, wherein the primary goal is to develop broad plan based on representative rather than exact information, the level of precision offered by the conventional approach becomes redundant. To this end, continuous approximation (CA) method offers a practical alternative with use of continuous density functions that estimate parameters approximately (Ansari et al., 2018). In particular, by abstracting discrete information into continuous function forms, this approach strikes a balance between estimation accuracy and computational effort, facilitating agile and efficient decision-making. Nonetheless, CA-based routing frameworks have a limited scope as they assume a simplified distribution environment (Franceschetti et al., 2017). Specifically, these frameworks consider a logistic operator catering to customers having uniformly dispersed random locations and fixed unitary demand from a distribution facility located sufficiently far from the service region. This assumption significantly limits the capability of such frameworks to accurately model last-mile operations in real-world distribution environments. For instance, postal service, e-commerce, q-commerce, waste collection, emergency service, etc., have peculiar distribution structure and distinctive customer characteristics. Such heterogeneity in distribution environments therefore necessitates a more comprehensive framework.

Hence, the objective of this work is to develop a holistic CA framework for the Vehicle Routing Problem (VRP) to facilitate strategic last-mile distribution planning within diverse logistic settings. To do so, this study a) synthesizes a range of distribution environments, each with a specific facility location, unique spatial dispersion, and distinct demand pattern; it then b) optimizes last-mile operations for each distribution environment instance using an Adaptive Large Neighborhood Search (ALNS) metaheuristic; and consequently, this work c) develops a robust CA-based functional form through symbolic regression to precisely estimate the total tour length across the various distribution environments. Note, to establish reliability of this work, the study validates the proposed model against benchmark VRP instances and subsequently compares its performance against existing CA-based models.

Overall, this work significantly advances the state-of-the-art in CA-based routing frameworks. By abstracting complex, discrete information into continuous functional form, this study provides an efficient alternative to addressing the VRP, especially, within the context of strategic planning. Specifically, through robust approximation of delivery tour length across diverse logistic settings, each with a peculiar distribution structure and distinctive customer characteristics, this study enables estimation of operational costs in the absence of a precise operational plan. Hence, the contributions of this work support strategic decision-making for a wide scope of logistic operators that may have representative but not necessarily exact information pertaining to the distribution environment.

This work is organized as follows: Section 2 provides a comprehensive review of extant literature on CA-based routing frameworks, outlining principles of CA in the context of last-mile distribution modelling, tracing model development through the years, and identifying critical research gaps, thereby establishing the novelty and contributions of this work. Section 3 focuses on synthesizing real-world distribution environments in the form of VRP instances, thus addressing the first objective of this study. Section 4 concerns with the second objective, detailing the ALNS metaheuristic deployed in this work to optimize the last-mile operations for each instance. Section 5 follows up with the final objective, rendering a thoroughly validated and tested CA-based functional form for total distribution tour length. Finally, this study concludes with a section consolidating the key findings of this work to discuss critical managerial implications, address practical limitations, and propose future research scope in this context.

2. Literature review

Typical CA-based routing frameworks for VRP consider a simplified distribution environment (Franceschetti et al., 2017). Here, a logistic operator must serve n identical customers located randomly, uniformly, and independently across a service region of size a, with a homogenous fleet of delivery vehicles stationed at a distribution facility located sufficiently far from the service region. For this logistic operator, a typical delivery tour comprises of the long-haul – journey from the distribution facility to the first customer, and likewise, from the last customer to the distribution facility, as well as the last-mile – journey between the first and the last customer. Hence, the CA-based routing frameworks estimate the length of this delivery tour as the sum of the back-and-forth long-haul distance and the last-mile distance. Specifically, as established in

the seminal work of Daganzo (1984), the long-haul is given by the average distance between the distribution facility and the customers – ρ, while the last-mile is continuously approximated, proportional to the number of customers in the tour – c, and inversely proportional to the square root of customer density – $\delta = n/a$ (equation 1).

$$l = 2\rho + k\frac{c}{\sqrt{\delta}} \tag{1}$$

Building on this foundational work, Figliozzi (2008, 2009, 2010) expanded the CA frameworks for VRP with dimensionless coefficients to incorporate the impact of demand level, time-window, and congestion on delivery tour length. On the other hand, Nicola et al. (2019) formulated regression-based approximations to account for specific facility locations and unique customer spatial distributions. Similar advancements can also be traced for CA models for Travelling Salesman Problem (TSP). For instance, Merchán and Winkenbach (2019) introduced a circuity factor to address the oversimplification of rectilinear distances in CA frameworks for TSP, thus better reflecting urban road networks. Moreover, Çavdar and Sokol (2015) introduced a distribution-free model incorporating spatial dispersion indicators to approximate the length of a TSP. Similarly, Rifki et al. (2021) and Rifki and Garaix (2023) developed asymptotic tour length approximations for varying scenarios of spatial and temporal demand distributions of customers. These contributions have enabled researchers to integrate CA models across multiple decision layers, particularly through the aggregation of cost structures from facility location, routing, and inventory management into unified objective functions, yielding closed-form equations (Davis and Figliozzi, 2013; Estrada and Roca-Riu, 2017; Ghaffarinasab et al., 2018; Ouyang and Daganzo, 2006; Pahwa and Jaller, 2022; Saberi and Verbas, 2012; Turkensteen and Klose, 2012). For the most recent review of advancements in CA frameworks for operations research, refer to Ansari et al. (2018).

As mentioned earlier, these CA-based routing frameworks consider a simplified distribution environment with a logistic operator catering to customers having uniformly dispersed random locations and fixed unitary demand from a distribution facility located sufficiently far from the service region. However, real-world distribution environments may in fact have a specific facility location (ρ_x, ρ_y), unique spatial dispersion (f_X, f_Y, f_{XY}), and distinct demand pattern (f_Q) (Ansari et al., 2018; Franceschetti et al., 2017; Langevin et al., 1996). For such environments, equation 2 more generally expresses the length of delivery tour. Here, the first term evaluates the long-haul distance (Pahwa and Jaller, 2022), while the second term estimates the last-mile distance (Beardwood et al., 1959). Consequently, for the simplified distribution environment, equation 2 simplifies into equation 1. However, for real-world distribution environments, this equation cannot be resolved into a closed form.

$$l = 2\left[\int_X |x - \rho_x| f_X(x)dx + \int_Y |y - \rho_y| f_Y(y)dy\right] + \beta\left[\iint_{XY} \sqrt{f_{XY}(x,y)}\, dxdy\right]\frac{c}{\sqrt{\delta}} \tag{2}$$

Thus, a holistic CA-based routing framework that can enable strategic decision-making within diverse logistic settings makes for a novel contribution to the field of operations research. To the authors' knowledge only Nicola et al. (2019) have explored application of CA models for realistic distribution environments in the context of VRP. However, the proposed CA-based VRP models in the study suffer from poor interpretability owing to use to rigid functional forms, as well as are susceptible to overfitting due to lack of thorough validation. To address this critical gap in the literature, this study employs form-agnostic symbolic regression approach to develop a robust CA-based functional form for VRP that can precisely estimate the total tour length across the various distribution environments, each with a peculiar distribution structure and distinctive customer characteristics. Consequently, this enables estimation of operational costs in the absence of a precise operational plan, thus facilitating long-term strategic planning for a wide range of logistic operators that may have representative but not necessarily exact information pertaining to the distribution environment.

3. Distribution environments

To develop a holistic CA-based routing framework, this study synthesis a total of 1080 distribution environments of size $l_x \times l_y$; $(l_x, l_y \in [1, 9])$, wherein a logistic operator must cater to n; $(n \in [100, 500])$ customers that have f_s; $(f_s \in \{U_s, T_s, NT_s, ET_s, C_s, B_s\})$ spatial dispersion and f_d; $(f_d \in \{U_d, T_d, NT_d, ET_d, C_d, B_d\})$ demand pattern, using a homogenous fleet of delivery vehicles, each with capacity q_v; $(q_v \in \{z \in \mathbb{Z}; 10 \le z \le 100\})$, stationed at a distribution facility located at (ρ_x, ρ_y); $(\rho_x/l_x, \rho_y/l_y \in \{z = -1 + 0.1k; k \in \mathbb{Z}, -1 \le z \le 1\})$ relative to the center of this service region. Note that in this study, the standard unit of measure (UoM) for coordinates pertains to distance, such as kilometers or miles, depending on the regional context. Likewise, the standard UoM for customer demand may represent quantity, weight, or volume of packages, depending on the logistics setting.

Here, U_s refers to Uniform Spatial Distribution, wherein customer locations follow uniform distribution between the boundaries of the service region, much like conventional CA-based routing frameworks. Further, as the name suggests, in Triangular Spatial Distribution (T_s), customers' horizontal and vertical coordinates follow a triangular distribution – $T(a, b, c)$, wherein limits a and b represent the service region boundaries, while mode c, drawn from the uniform distribution – $U(a, b)$, determines customer agglomeration. On the other hand, NT_s refers to Normal (Truncated) Spatial Distribution, wherein customers' coordinates follow normal distribution with mean μ and variance σ drawn from the uniform distributions – $U(a, b)$ and $U(0.2|a|, 0.8|b|)$, respectively, truncated between limits a and b defined by the boundaries of the service region. Likewise, in Exponential (Truncated) Spatial Distribution (ET_s), customers' x and y coordinates follow exponential distribution with rate parameter θ drawn from the uniform distribution $U(0, |a| + |b|)$, truncated between limits a and b defined by the service region boundaries. However, under Constant Spatial Distribution (C_s), customers are distributed in a $n_x \times n_y$ (least differentiated integer factors of n) grid. And, in B_s, i.e., Boundary Spatial Distribution, customers are located near service region boundaries such that $|x_i|/0.5l_x^* |y_i|/0.5l_y > U(0, 1) \forall i \in [1, n]$. Note, Table 1 outlines the various customer dispersions considered in this work, detailing their underlying mathematical formulations that define customers' coordinates (relative to the center of the service region in standard units of measure), while along Fig. 1 presents illustrative instances for the same.

In the context of customer demand, U_d refers to Uniform Demand Pattern, wherein customer demand follows uniform distribution between UoM limits a and b. On the other hand, in Triangular Demand Pattern (T_d), customer demand follows a triangular distribution – $\text{Triangular}(a, b, c)$, wherein limits a and b represent the UoM limits, while mode c is drawn from the uniform distribution – $\text{Uniform}(a, b)$. Further, in Normal (Truncated) Demand Pattern (NT_d), customer demand follows normal distribution with mean μ and variance σ drawn from the uniform distributions – $\text{Uniform}(a, b)$ and $\text{Uniform}(0.2|a|, 0.8|b|)$, respectively, truncated between UoM limits a and b. Similarly, in Exponential (Truncated) Demand Pattern (ET_d), customer demand follows exponential distribution with rate parameter θ drawn from the uniform distribution $\text{Uniform}(0, |a| + |b|)$, truncated between UoM limits a and b. In Constant

Table 1
Customer Spatial Distribution – Formulation.

Spatial Distribution	Formulation
Uniform	$\{\mathrm{U}(-0.5l_k, 0.5l_k); i \in [1,n], k \in \{x,y\}\}$
Triangular	$\{\mathrm{T}(-0.5l_k,\ 0.5l_k, \mathrm{U}(-0.5l_k,\ 0.5l_k))\ ; i \in [1,n], k \in \{x,y\}\}$
Normal (Truncated)	$\{NT(\mathrm{U}(-0.5l_k, 0.5l_k), \mathrm{U}(0.1l_k, 0.4l_k), -0.5l_k, 0.5l_k); i \in [1,n], k \in \{x,y\}\}$
Exponential (Truncated)	$\{\mathrm{ET}(\mathrm{U}(0,l_k), 0, l_k) - 0.5l_{k\mathrm{OR}} 0.5l_k - \mathrm{ET}(\mathrm{U}(0,l_k), 0, l_k); i \in [1,n], k \in \{x,y\}\}$
Constant	$\{(-0.5 + (i-1)/(n_k - 1))l_x; i \in [1,n_k], k \in \{x,y\}\}$
Boundary	$\{\mathrm{U}(-0.5l_k, 0.5l_k); i \in [1,n], k \in \{x,y\}\}; \lvert x_i\rvert/0.5l_x{}^* \lvert y_i\rvert/0.5l_y > \mathrm{U}(0,1) \forall i \in [1,n]$

U: Uniform, T: Triangular, NT: Normal (Truncated), ET: Exponential (Truncated), C: Constant, B: Boundary.

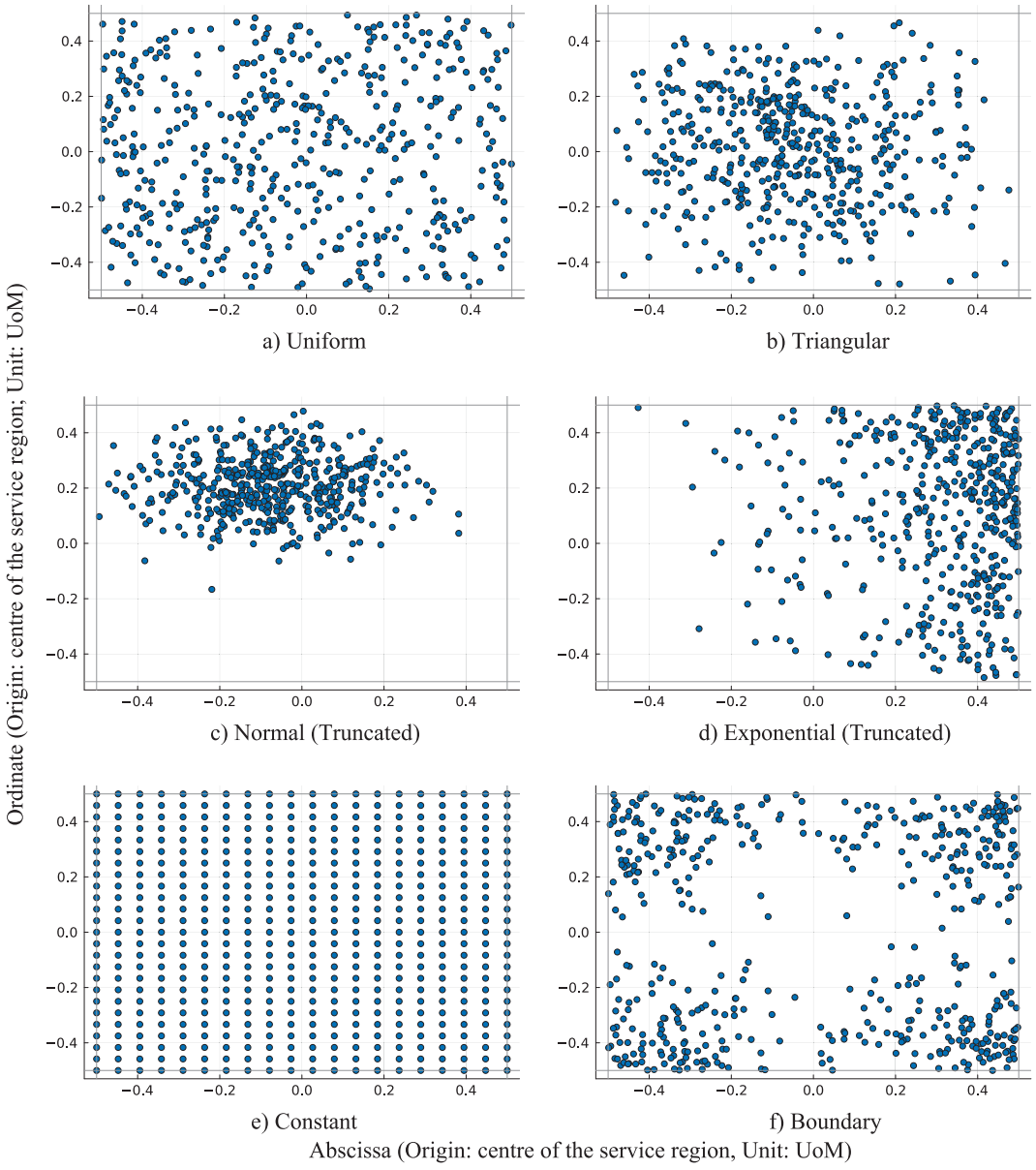


Fig. 1. Customer Spatial Dispersion – Illustrative Instances.

Table 2
Customer Demand Pattern– Formulation.

Demand Pattern	Formulation
Uniform	$\{U(0,2); i \in [1,n]\}$
Triangular	$\{T(0,2,U(0,2)); i \in [1,n]\}$
Normal (Truncated)	$\{NT(U(0,2),U(0.2,0.8),0,2); i \in [1,n]\}$
Exponential (Truncated)	$\{ET(U(0,2),0,2)_{OR} 2 - ET(U(0,2),0,2); i \in [1,n]\}$
Constant	$\{1; i \in [1,n]\}$
Boundary	$\{U(0,2); i \in [1,n]\}; \lvert q_i - 1\rvert > U(0,1) \forall i \in [1,n]$

U: Uniform, T: Triangular, NT: Normal (Truncated), ET: Exponential (Truncated), C: Constant, B: Boundary.

Demand Pattern (C_d), customers have a unit demand, much like conventional CA-based routing frameworks. And in Boundary Demand Pattern (B_d), customer demand is concentrated near the UoM limits. Note, Table 2 describes the different demand patterns assumed in this work, detailing the underlying mathematical formulations that define customers' demand (in standard units of measure) and Fig. 2 presents the associated probability density functions for the same.

4. Adaptive Large Neighborhood search metaheuristic

The Adaptive Large Neighborhood Search (ALNS) metaheuristic *searches* through the *neighborhood* by destroying and consequently rebuilding the solution thereby reconfiguring *large* portions of the solution using specific operators that are chosen *adaptively* in each iteration of the algorithm based on the performance of operators in the previous iterations, hence the name adaptive large neighborhood search (Ropke and Pisinger, 2006). Below is a detailed description of the framework of this algorithm along with its implementation (and validation), tailored towards generating high-quality solutions for the synthetic distribution environments described in the previous section.

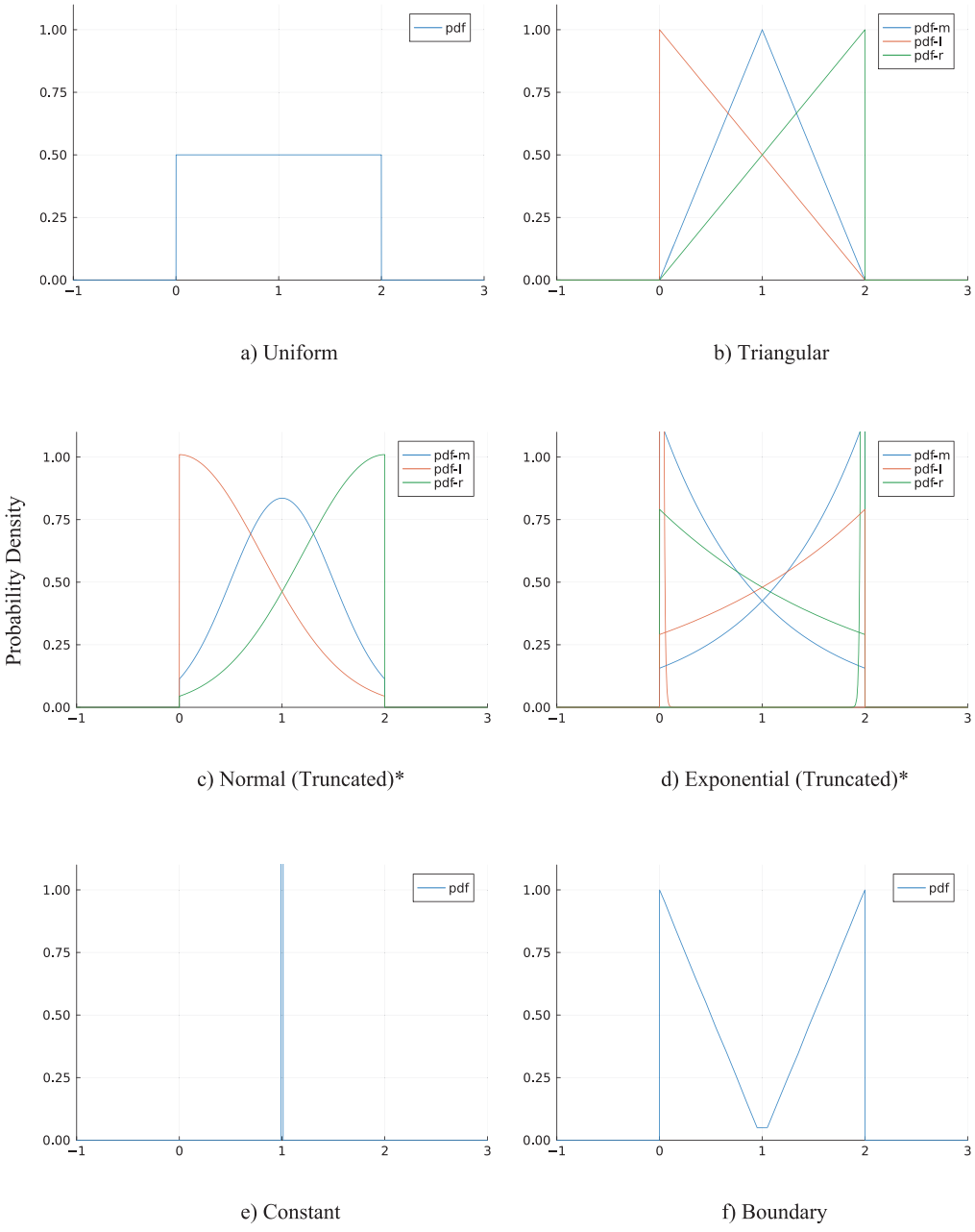


Fig. 2. Customer Demand Pattern – Probability Density.

Algorithm
Adaptive Large Neighborhood Search

Procedure ALNS($s_o, (j, k, n, m, \boldsymbol{o_r}, \boldsymbol{o_i}, \boldsymbol{o_l}, \sigma_1, \sigma_2, \sigma_3, \underline{e}, \overline{e}, \underline{\mu}, \overline{\mu}, \underline{\omega}, \overline{\omega}, \underline{\tau}, \overline{\tau}, \varphi, \rho)$)
$s \leftarrow s_o$ // initialise current solution $-s$ as the initial solution $-s_o$
$s^* \leftarrow s$ //initialise best solution $-s^*$ as the current solution
$H \leftarrow \{h(s)\}$ // initialise hash list
$T \leftarrow \overline{\omega} f(s^*)/\ln(1/\overline{\tau})$ // initialise the current temperature based on the cooling schedule
for $o_r \in \boldsymbol{o_r}$ **do** // initialise removal operator weights to 1
  $w_r \leftarrow 1$
end for
for $o_i \in \boldsymbol{o_i}$ **do** // initialise insertion operator weights to 1
  $w_i \leftarrow 1$
end for
$u \leftarrow 1$ //initialise segment index to 1
while $u \leq j$ **do** //repeat for j segments
  for $o_r \in \boldsymbol{o_r}$ **do**
    $p_r \leftarrow w_r / \sum_{r \in \Psi_r} w_r$ // update removal operator probability
  end for
  for $o_i \in \boldsymbol{o_i}$ **do**
    $p_i \leftarrow w_i / \sum_{r \in \Psi_i} w_i$ // update insertion operator probability
  end for
  for $o_r \in \boldsymbol{o_r}$ **do**
    $c_r \leftarrow 0$ // set removal operator count to 0
    $\pi_r \leftarrow 0$ // set removal operator score to 0
  end for
  for $o_i \in \boldsymbol{o_i}$ **do**
    $c_i \leftarrow 0$ // set insertion operator count to 0
    $\pi_i \leftarrow 0$ // set insertion operator score to 0
  end for
  $v \leftarrow 1$ //initialise iteration index to 1
  while $v \leq n$ **do** //repeat for n iterations
    $o_r \overset{R_{p_r}}{\leftarrow} \boldsymbol{o_r}$ // randomly select a removal operator
    $o_i \overset{R_{p_i}}{\leftarrow} \boldsymbol{o_i}$ // randomly select an insertion operator
    $c_r \leftarrow c_r + 1$ // update removal operator count
    $c_i \leftarrow c_i + 1$ // update insertion operator count
    $\Lambda \sim U(0,1)$
    $\lambda \overset{R}{\leftarrow} \Lambda$
    $q \leftarrow \left\lceil \begin{matrix} (1-\lambda)\min(\underline{e}, \underline{\mu}\|s\|) \\ +\lambda \min(\overline{e}, \overline{\mu}\|s\|) \end{matrix} \right\rceil^-$ // set the size of removal and insertion operation
    $s' \leftarrow o_i(o_r(q, s))$ // remove and insert select number of customer nodes
    if $f(s') < f(s^*)$ **then** // if the new solution is better than the best solution then
      $s^* \leftarrow s'$ // update the best solution to the current solution
      $s \leftarrow s'$ //update the current solution to the new solution
      $\pi_r \leftarrow \pi_r + \sigma_1$ // update removal operator score by σ_1
      $\pi_i \leftarrow \pi_i + \sigma_1$ // update insertion operator score by σ_1
    else if $f(s') < f(s)$ **then** // else if the new solution is better than the current solution then
      $s \leftarrow s'$ //update the current solution to the new solution
      if $h(s) \notin H$ **then** // if the solution does not exist in the hashed tabu list
        $\pi_r \leftarrow \pi_r + \sigma_2$ // update removal operator score by σ_2
        $\pi_i \leftarrow \pi_i + \sigma_2$ // update insertion operator score by σ_2
      end if
    else //else accept new solution with a small probability
      $\Lambda \sim U(0,1)$
      $\lambda \overset{R}{\leftarrow} \Lambda$
      if $\lambda < \exp\left(\frac{f(s) - f(s')}{T}\right)$ **then**
        $s \leftarrow s'$ //update the current solution to the new solution
        if $h(s) \notin H$ **then** // if the solution does not exist in the hashed tabu list
          $\pi_r \leftarrow \pi_r + \sigma_3$ // update removal operator score by σ_3
          $\pi_i \leftarrow \pi_i + \sigma_3$ // update insertion operator score by σ_3
        end if
      end if
    end if
    $H \leftarrow H \cup \{h(s)\}$ //add the current solution to the hash list
    $T \leftarrow \max(\varphi T, \underline{\omega} f(s^*)/\ln(1/\underline{\tau}))$ //update current temperature based on the cooling schedule
    $v \leftarrow v + 1$ //update iteration index
  end while
  if $h \bmod k$ // after every k segments reset best current solution to best solution
    $s \leftarrow s^*$
  end if
  for $o_r \in \boldsymbol{o_r}$ **do** // update removal operator weights
    if $c_r \neq 0$ **then**
      $w_r \leftarrow \rho \pi_r / c_r + (1 - \rho) w_r$
    end if
  end for

(*continued on next page*)

Algorithm (*continued*)

```
72 :  for o_i ∈ o_i do                                  // update insertion operator weights
73 :    if c_i ≠ 0 then
74 :      w_i ← ρπ_i/c_i + (1 − ρ)w_i
75 :    end if
76 :  end for
77 :  for o_l ∈ o_l                                     // iteratively perform local search on the current solution
78 :    s ← o_l(s, m)
79 :  end for
80 :  if f(s) < f(s*) then                              // if the current solution is better than the best solution then
81 :    s* ← s                                          // update the best solution to the current solution
82 :  end if
83 :  H ← H ∪ {h(s)}                                    //add the current solution to the hash list
84 :  u ← u + 1                                        //update segment index
85 :  end while
86 :  return s*                                        // return the best solution
```

4.1. Framework

The implementation in this work follows the standard ALNS structure. It maintains a current solution – s, and tracks the best solution found so far – s^*, initializing both from a starting configuration – s_o. The core search proceeds iteratively over j segments, each comprising n individual iterations. The fundamental operation within each iteration is a ruin-and-recreate cycle. This begins with the selection of a removal and an insertion operator – o_r and o_i, from predefined sets $\boldsymbol{o_r}$ and $\boldsymbol{o_i}$, respectively. This selection is guided by a roulette wheel mechanism employing adaptive weights that reflect the recent historical performance of each operator ($w_r; o_r \in \boldsymbol{o_r}$ and $w_i; o_i \in \boldsymbol{o_i}$). The chosen removal operator then partially destroys the current solution, removing a certain quantity of elements (ranging between $\underline{e}$ and $\overline{e}$) or fraction of elements (ranging between $\underline{\mu}$ and $\overline{\mu}$). Following the ruin phase, the selected insertion operator rebuilds the solution, generating a new candidate solution s'.

A key aspect of the ALNS methodology is its adaptive learning mechanism for operator selection. The algorithm monitors operator success within each segment using scores ($\pi_r; o_r \in \boldsymbol{o_r}$ and $\pi_i; o_i \in \boldsymbol{o_i}$), which are reset at the start of every segment. In each iteration, the operator scores are updated based on the quality and uniqueness of the solutions generated. Subsequently, at the conclusion of a segment, the weights w_r and w_i are updated based on these accumulated scores, operator usage counts ($c_r; o_r \in \boldsymbol{o_r}$ and $c_i; o_i \in \boldsymbol{o_i}$), and a reaction factor – ρ. This factor determines how strongly recent performance influences the weights, while a dissipation factor of $(1-\rho)$ provides stability by retaining some influence from previous segments. The recalculated weights then guide operator selection in the subsequent segment.The acceptance of the newly generated solution s' into the search process depends on its objective function value f relative to the current solution s and the best-known solution s^*, potentially considering solution novelty determined via a hash function h. If s' yields a better objective value than s^* ($f(s') < f(s^*)$), it is unconditionally accepted, replacing both s and s^*, and the operators used receive the highest reward, amounting to σ_1. If s' improves only upon the current solution ($f(s^*) \leq f(s') < f(s)$), it replaces s, and the operators earn a standard reward of σ_2. To facilitate escape from local optima, the framework also incorporates a mechanism, akin to Simulated Annealing, to accept non-improving solutions

Table 3
ALNS parameters calibrated to VRP.

ALNS Parameters		Value
Number of segments	j	50
Number of segments to reset the solution	k	5
Number of iterations in a segment	n	n
Number of iterations in a local search	m	100 n
Removal operators	$\boldsymbol{o_r}$	random-customer random-route random-vehicle related-customer related-route related-vehicle worst-customer worst-route worst-vehicle
Insertion operators	$\boldsymbol{o_i}$	random greedy-deterministic greedy-stochastic regret-2 regret-3
Local search operators	$\boldsymbol{o_l}$	intra-move intra-swap intra-opt inter-move inter-swap inter-opt
Reward for a new best solution	σ_1	15
Reward for a new better solution	σ_2	10
Reward for a new worse but accepted solution	σ_3	3
Minimum elements removed	$\underline{e}$	4
Maximum elements removed	$\overline{e}$	60
Minimum removal fraction	$\underline{\mu}$	0.1
Maximum removal fraction	$\overline{\mu}$	0.4
Initial temperature deviation parameter	$\overline{\omega}$	5 %
Initial temperature probability parameter	$\overline{\tau}$	50 %
Final temperature deviation parameter	$\underline{\omega}$	1 %
Final temperature probability parameter	$\underline{\tau}$	1 %
Cooling rate	φ	0.9985
Reaction factor	ρ	0.1

Table 4
Summary of Benchmark Datasets Used for Validation.

Dataset Source	Instances	Size
Augerat (1995)	A-n36-k5, A-n45-k6, B-n50-k7, A-n53-k7, B-n56-k7, B-n66-k9, A-n80-*k*10	35–79
Christofides and Eilon (1969)	E-n22-k4, E-n51-k5, E-n76-k14, E-n101-k14	21–100
Christofides et al. (1979)	CMT4	65
Fisher (1994)	F-n135-k7	134
Golden et al. (1998)	Golden_8, Golden_9, Golden_12, Golden_15, Golden_16	255–483
Rochat and Taillard (1995)	tai385	385
Uchoa et al. (2017)	X-n115-*k*10, X-n125-k30, X-n143-k7, X-n176-k26, X-n186-k15, X-n195-k51, X-n204-k19, X-n219-k73, X-n228-k23, X-n237-k14, X-n247-k50, X-n266-k58, X-n275-k28, X-n289-k60, X-n298-k31, X-n308-k13, X-n317-k53, X-n327-k20, X-n336-k84, X-n344-k43, X-n359-k29, X-n367-k17, X-n376-k94, X-n401-k29, X-n420-k130, X-n429-k61, X-n439-k37, X-n459-k26, X-n469-k138, X-n480-k70, X-n502-k39	114–501

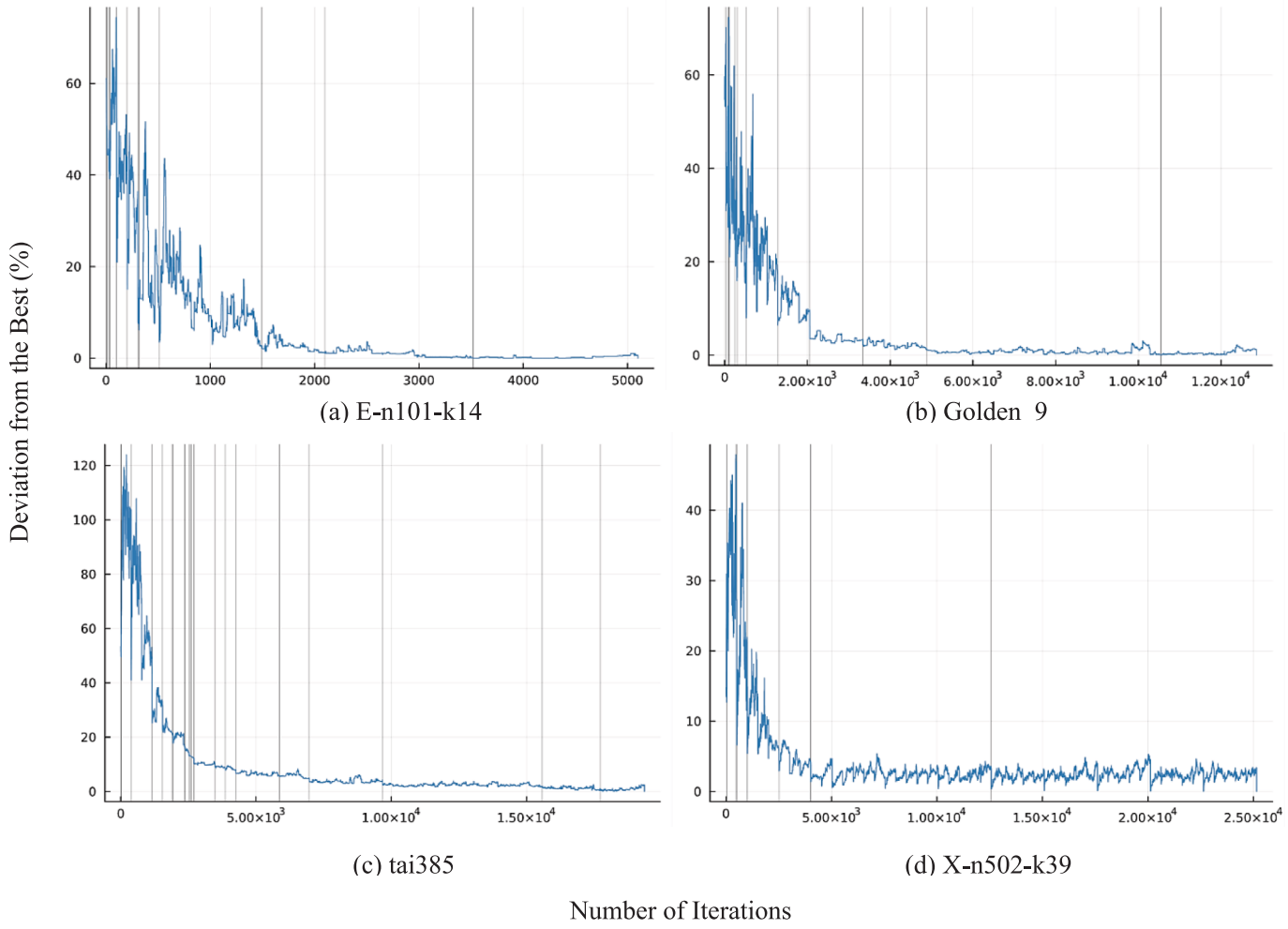


Fig. 3. Illustrative Convergence Plots.

Table 5
ALNS validation – Deviation from the best – Quartiles.

Quartile	Gap		
	Min	Mean	Max
Min	0.1 %	0.4 %	0.4 %
1st Quartile	0.9 %	2.0 %	2.8 %
Median	1.7 %	2.9 %	4.0 %
3rd Quartile	2.9 %	3.6 %	5.4 %
Max	5.0 %	6.4 %	7.9 %

$(f(s') > f(s))$. Such solutions are accepted probabilistically based on the Boltzmann criterion, $\lambda < \exp(f(s) - f(s')/T_k)$(where $\lambda \in [0,1]$ is random), with acceptance more likely at higher temperatures – T_k. Operators leading to such accepted solutions are given a smaller reward of σ_3.

The algorithm initializes this temperature at a level designed to permit acceptance of an $\overline{\omega}$ worse solution, with a target probability $\overline{\tau}$. In each subsequent iteration, the temperature is reduced by a cooling factor φ, progressively decreasing the likelihood of accepting inferior solutions, to a minimum that would still enable the algorithm to accept an $\underline{\omega}$ worse solution with a $\underline{\tau}$ probability. To prevent stagnation, the search is periodically refocused by resetting the current solution s to the best-found solution s^* every k segments. Additionally, solution quality is further refined through an optional local search phase applied at the end of each segment, running for m iterations using operators from a set $\boldsymbol{o}_l$. After completing the predefined n segments, the ALNS procedure terminates, returning the best solution s^* identified.

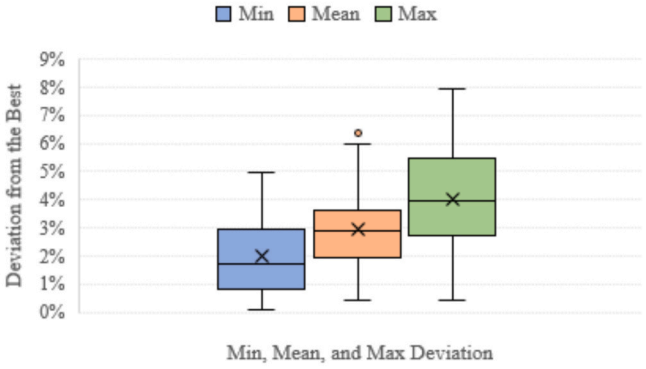


Fig. 4. Box Plot – Deviation from the Best.

For a detailed step-by-step implementation, the reader is referred to the pseudocode presented below.

4.2. Implementation

This study employs Julia v1.9.4 (Bezanson et al., 2017) on an Intel (R) Xeon(R) Platinum 8160 CPU with 768 GB RAM and two processors each at 2.10 GHz, to deploy metaheuristic optimization framework.

Table 6
Variation in distribution length across different facility locations, customer dispersions, and demand patterns with respect to uniform customer dispersion and uniform demand pattern.

a) Facility Location – Inside the Service Region

Distribution Length (km)		Demand Pattern					
		Uniform	Triangular	Normal(Truncated)	Exponential(Truncated)	Constant	Boundary
Spatial Dispersion	Uniform	94.94	103.54	111.55	119.80	94.23	96.41
	Triangular	73.94	79.78	84.04	90.80	73.44	73.74
	Normal(Truncated)	67.56	74.91	79.61	87.76	67.29	68.13
	Exponential(Truncated)	85.96	93.14	98.97	105.47	84.64	85.73
	Constant	91.64	100.41	107.18	102.55	90.99	91.38
	Boundary	96.78	105.86	113.57	122.81	95.35	97.39

b) Facility Location – Outside the Service Region

Distribution Length (km)		Demand Pattern					
		Uniform	Triangular	Normal(Truncated)	Exponential(Truncated)	Constant	Boundary
Spatial Dispersion	Uniform	155.38	150.22	145.38	165.82	153.30	155.20
	Triangular	148.94	144.53	138.75	158.16	148.11	148.85
	Normal(Truncated)	137.12	134.40	129.78	143.16	136.52	137.74
	Exponential(Truncated)	125.32	122.23	117.49	137.47	124.42	125.21
	Constant	157.01	152.48	146.29	168.74	155.54	157.20
	Boundary	152.16	148.54	143.84	163.93	151.46	153.65

Specifically, this work tailors Pahwa (2023) implementation of the ALNS metaheuristic to develop high-quality solutions for the Vehicle Routing Problem (VRP) (Table 3).

Given a VRP instance with n number of customers, this study provides the ALNS algorithm with an initial solution developed using an iterated k-means clustering algorithm. Starting with a single cluster, this algorithm progressively increases the size of the cluster to as many clusters as the number of customers, generating a solution in each iteration until it finds a feasible solution.

The ALNS algorithm then proceeds to run 50 segments, each comprising of as many iterations as the size of the instance. Within each such iteration, the algorithm employs a roulette wheel selection procedure to randomly select a removal and an insertion operator from a set of nine removal and five insertion operators, respectively. The nine removal operators combine of one of the three removal principles – random, related, and worst, with one of the three removal scopes – customer, route, and vehicle, inspired from the works of Ropke and Pisinger (2006), Contardo et al. (2012), Azi et al. (2014), and Grangier et al. (2016). While the five insertion operators include random, greedy-deterministic, greedy-stochastic, regret-2 and regret-3 insertion, motivated from the works of Ropke and Pisinger (2006) and Contardo et al. (2012). Note, the selected removal operator removes a certain random number of customer nodes from the current solution, ranging between a minimum of min(4, 0.1 n) and a maximum of min(60, 0.4 n) customer nodes. Subsequently, the selected insertion operator re-inserts these customer nodes into the partial solution.

Upon generation of a new solution, the ALNS algorithm performs specific updates based on the quality the new solution evaluated by an objective function that sums the total cost of routing along with any penalties stemming from constraint violations (applied proportionally to the magnitude of violation), as well as its novelty determined by a hash function that encodes the solution based on the sequence of customer visits. In particular, if the resulting new solution is better than the best solution, then the algorithm updates both, the current as well as the best solution, and subsequently rewards the operators with a score of 15. However, if the new solution is only better than the current solution, then the algorithm updates the current solution, while rewarding the operators with a smaller score of 10, yet only if the new solution is also novel. Nonetheless, if the new solution is worse than the current solution, then the algorithm accepts it as the current solution with a small probability, and consequently, the operators receive an even smaller reward of 3 for novelty.

Note, the algorithm accepts a worse new solution through a search temperature initialized to enable the ALNS algorithm to accept a 5 % worse solution with a 50 % probability. In each iteration thereafter, the temperature drops by a factor of 0.9985, to a minimum that would still enable the algorithm to accept 1 % worse solution with a 1 % probability. However, to prevent the algorithm from incessantly exploring poor search space from repeated acceptance of worse solutions, the current solution is reset to the best solution, periodically, after every 5 segments.

At the end of each segment, the algorithm further refines the current solution through six local search operators, applied iteratively, each for 100 n iterations. Notably, the six local search operators combine an intra-route or inter-route search scope with one of the three search principles – move, swap, and opt.

Subsequently, the algorithm updates operator weights accounting for operator performance from the current segment, weighted by a reaction factor of 0.1, while retaining operator performance from previous segments via a dissipation factor of 0.9. Finally, after 50 n iterations (n iterations performed in a batch of 50 segments each), the algorithm terminates, returning the best-found solution.

4.3. Validation

This work validates the performance of the proposed ALNS algorithm on a diverse collection of 50 benchmark instances drawn from several well-established VRP datasets in the literature (Table 4). Fig. 3 presents convergence plots of the algorithm on a few select instances. To comprehensively assess algorithm's consistency, this study solved each of the 50 instances 10 times for distinct seeds of the Mersenne Twister random number generator (1010, 1106, 1509, 1604, 1905, 2104, 2412, 2703, 2710, and 2807). Consequently, the validation process renders Table 5, representing quartiles of minimum, mean, and maximum deviation from the best across all instances, while Fig. 4 represents associated boxplots.

5. Continuous approximation

This section aims to formulate a closed-form, continuous approximation-based framework that can accurately estimate the delivery tour length corresponding to the near-optimal solutions of the synthetic distribution environments obtained using the Adaptive Large Neighborhood Search metaheuristic (Table 6), thereby supporting

strategic last-mile distribution planning for a logistic operator.

This study considers a logistic operator catering to n customers, wherein each customer $c \in C$ with demand d_c, located at (x_c, y_c) within a service region of size $a; (l_x \times l_y)$, is served from a distribution facility located at (ρ_x, ρ_y) relative to center of this service region, using a fleet of delivery vehicles, each with capacity of q_v. Note, a typical delivery tour for this logistic operator comprises of the long-haul – journey from the distribution facility to the first customer, and likewise, from the last customer to the distribution facility, as well as the last-mile – journey between the first and the last customer. Consequently, the CA-based routing framework in this study assumes long-haul length to be proportional to the average distanced between the distribution facility and customers – ρ, and the last-mile distance to be proportional to the nearest neighborhood distances – $s_c; c \in C$. Thus, under these assumptions, this study employs symbolic regression on optimized tour length values derived from the ALNS metaheuristic for synthetic distribution environments to develop closed-form, continuous approximation-based equations estimating the delivery tour length – l. Below is a list of notations and associated mathematical definitions of exogenous and endogenous variables employed for symbolic regression.

Note, symbolic regression is a machine learning technique that can discover underlying mathematical relationships between the endogenous and exogenous variables. Unlike the traditional regression methods, which fit data to specific model structure, symbolic regression explores an expansive space of mathematical expressions to uncover the suitable model structure. Due to its ability to identify interactions between endogenous and exogenous variables, symbolic regression offers tractable alternative to conventional machine learning methods, which focus on developing high-accuracy black-box models. Evidently, the advantage of symbolic regression lies in its ability to produce closed-form equations, facilitating interpretation, especially wherein understanding relationship between variables is as important as the prediction accuracy, as is the case in this study.

Definitions. *Exogenous variables* (*derived*).

$$n = |C| \tag{3}$$

$$a = (max\{x_c; c \in C\} - min\{x_c; c \in C\})(max\{y_c; c \in C\} - min\{y_c; c \in C\}) \tag{4}$$

$$\delta = n/a \tag{5}$$

$$s_c = \min\left\{\sqrt{(x_i - x_c)^2 + (y_i - y_c)^2}; i \in C \backslash \{c\}\right\} \tag{6}$$

Table 7
Descriptive Statistics – Absolute Percentage Error of F_1 Form Models on Training Set.

Measure	F_1C_0	F_1C_1	F_1C_2	F_1C_3	F_1C_4
Minimum	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %
1st quartile	1.7 %	1.4 %	1.5 %	1.5 %	1.4 %
Median	3.5 %	3.3 %	3.3 %	3.2 %	3.1 %
Mean	5.8 %	5.4 %	5.3 %	4.1 %	4.1 %
3rd quartile	7.1 %	6.3 %	6.2 %	5.7 %	5.8 %
Maximum	84.0 %	79.2 %	76.4 %	28.1 %	27.6 %

$$\rho_u = \frac{1}{n}\sum_{c \in C}\left((\rho_x - x_c)^2 + (\rho_y - y_c)^2\right)^{1/2} \tag{7}$$

$$\rho_w = \frac{1}{n}\sum_{c \in C} d_c\left((\rho_x - x_c)^2 + (\rho_y - y_c)^2\right)^{1/2} \tag{8}$$

$$\mu_z = \frac{1}{n}\sum_{c \in C} z_c; z = \{d, s\} \tag{9}$$

$$\sigma_z = \frac{1}{n}\left(\sum_{c \in C}(z_c - \mu_z)^2\right)^{1/2}; z = \{d, s\} \tag{10}$$

$$\gamma_z = \frac{1}{n}\left(\sum_{c \in C}(z_c - \mu_z)^3\right)^{1/3}; z = \{d, s\} \tag{11}$$

$$\kappa_z = \frac{1}{n}\left(\sum_{c \in C}(z_c - \mu_z)^4\right)^{1/4}; z = \{d, s\} \tag{12}$$

Endogenous variable

$$l = 2h(\mathbf{X}) + k(\mathbf{X})\frac{n}{\sqrt{\delta}} \tag{13}$$

Here, $h(\mathbf{X})$ is the long-haul length and $k(\mathbf{X})$ is the last-mile parameter.

5.1. Training Set

This subsection trains a total of 20 continuous approximation models over synthetic distribution environments using symbolic regression to accurately explain the variation in delivery tour length across facility locations, spatial dispersions, and demand patterns, depicted in Table 6. Notably, this table shows that conventional CA-based routing frameworks can yield prediction errors ranging from −27 % to 34 % when the distribution facility is located inside the service region, and −26 % to 8 % when located outside, owing to their underlying assumptions of

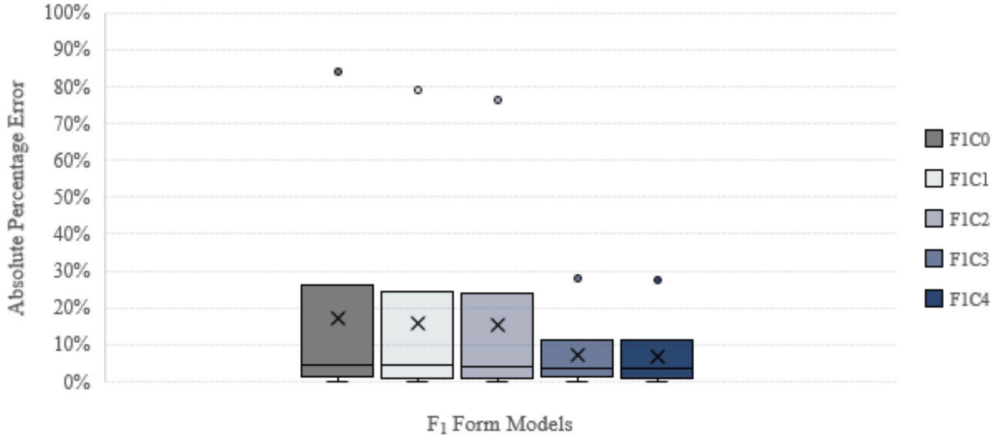


Fig. 5. Box Plot – Absolute Percentage Error of F_1 Form Models on Training Set.

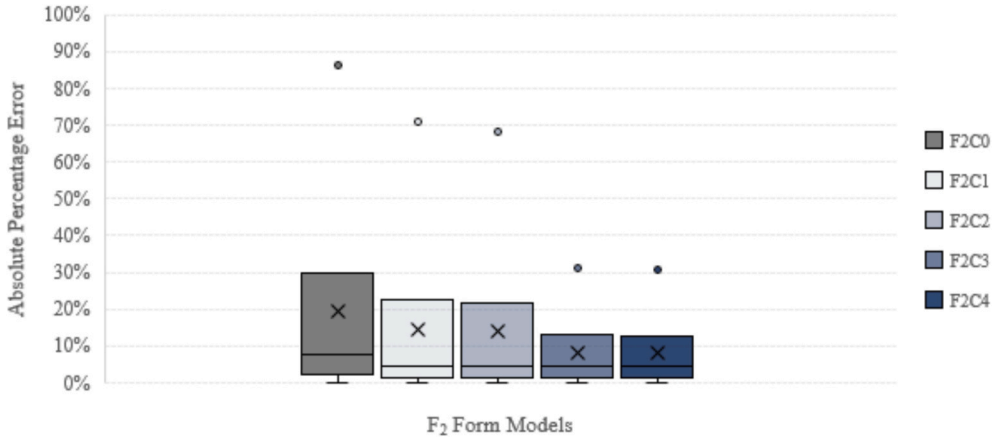


Fig. 6. Box Plot – Absolute Percentage Error (APE) of F_2 Form Models on Training Set.

Table 8
Descriptive Statistics – Absolute Percentage Error of F_2 Form Models on Training Set.

Measure	F_2C_0	F_2C_1	F_2C_2	F_2C_3	F_2C_4
Minimum	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %
1st quartile	2.9 %	1.5 %	1.6 %	1.9 %	1.8 %
Median	7.0 %	3.6 %	3.4 %	3.8 %	3.8 %
Mean	8.7 %	5.5 %	5.5 %	5.0 %	4.9 %
3rd quartile	11.2 %	6.4 %	6.3 %	6.9 %	6.9 %
Maximum	86.3 %	70.9 %	68.1 %	31.2 %	30.7 %

uniform customer spatial dispersion and constant customer demand pattern. This therefore underscores the need for a holistic CA framework for strategic last-mile distribution planning – the central objective of this study.

Note, the 20 models pertain to one of the four model forms $(F_i; i \in [1,4])$, each with five levels of varying model complexity $(C_j; j \in [0,4])$. Here, each model form assumes a specific structure for long-haul length – $h(\mathbf{X})$, while allowing symbolic regression to search for the structure associated with last-mile parameter – $k(\mathbf{X})$, with increasing levels of complexity. Specifically, the forms differ in terms of underlying assumptions regarding the distribution environment, incorporating unweighted/weighted distances and full-truckload/partial-truckload travel conditions. Furthermore, the complexity levels within each form progressively introduce additional statistical moments of customer spatial dispersion and demand pattern. Note, the discussion below details the complete structure for each of the 20 models along with their prediction accuracy on the training set using Absolute Percentage Error (APE) metric. This discussion is supported with boxplots that illustrate minimum and maximum values (lower and upper whisker), 1st and 3rd quartiles (bounds of the box), median value (line within the box), mean value ($\times$ symbol within the box), as well as outliers (if any; points beyond whiskers) of APE. In addition, these figures are supplemented with tables explicitly reporting the corresponding values of these measures.

F_1 Form: The F_1 form models (equations 14 – 18) assume long-haul length to be proportional to unweighted average distance between the distribution facility and the customers – ρ_u. Additionally, these models assume full-truckload travel, requiring the delivery fleet to fulfill the expected total customer demand in $n\mu_d/q_v$ delivery tours. As a result, the long-haul length is set to $\rho_u n\mu_d/q_v$, traversed to-and-fro by the delivery fleet. Subsequently, symbolic regression develops five models with increasing complexity $(C_j; j \in [0,4])$, with baseline model – F_1C_0 representing the model formulated by Daganzo (1984). Here, Fig. 5 and Table 7 describe prediction accuracy of these five models through APE.

Table 9
Descriptive Statistics – Absolute Percentage Error of F_3 Form Models on Training Set.

Measure	F_3C_0	F_3C_1	F_3C_2	F_3C_3	F_3C_4
Minimum	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %
1st quartile	1.6 %	1.5 %	1.4 %	1.5 %	1.4 %
Median	3.4 %	3.1 %	3.2 %	3.2 %	3.1 %
Mean	5.8 %	5.3 %	5.2 %	4.1 %	4.0 %
3rd quartile	7.0 %	6.2 %	6.2 %	5.7 %	5.8 %
Maximum	82.2 %	77.4 %	74.6 %	27.8 %	27.3 %

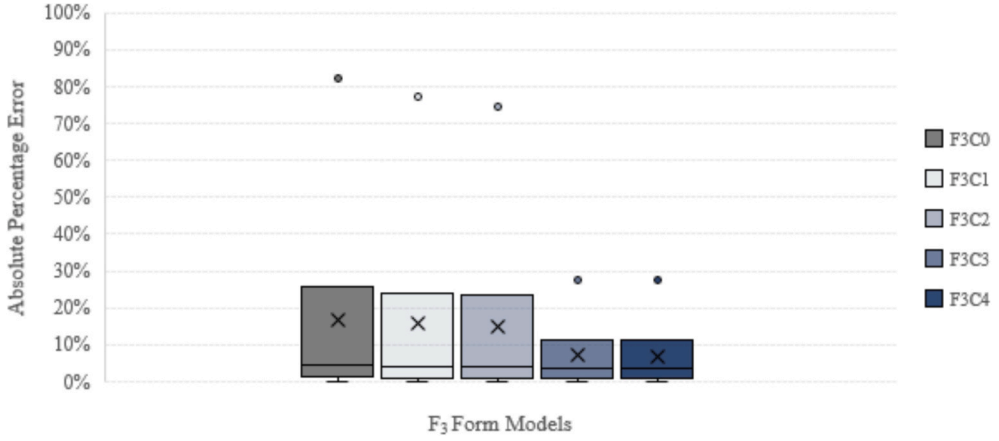


Fig. 7. Box Plot – Absolute Percentage Error of F_3 Form Models on Training Set.

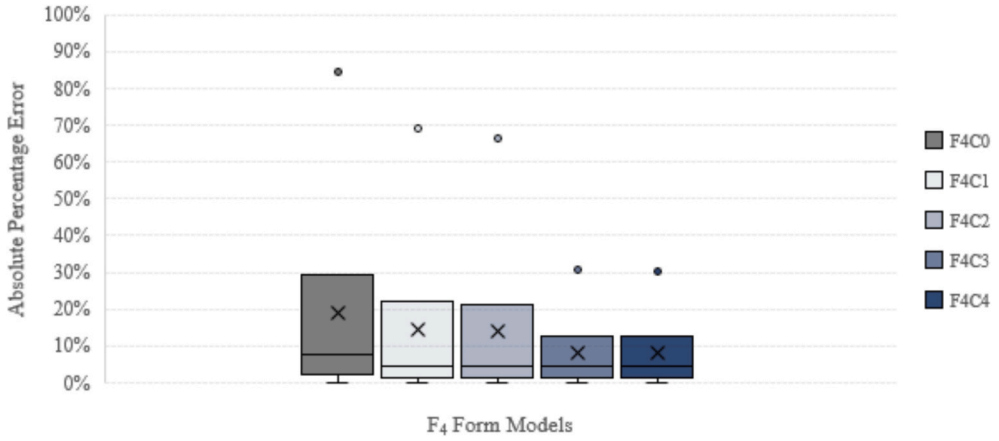


Fig. 8. Box Plot – Absolute Percentage Error of F_4 Form Models on Training Set.

Table 10
Descriptive Statistics – Absolute Percentage Error of F_4 Form Models on Training Set.

Measure	F_4C_0	F_4C_1	F_4C_2	F_4C_3	F_4C_4
Minimum	0.0 %	0.0 %	0.0 %	0.0 %	0.0 %
1st quartile	2.8 %	1.5 %	1.6 %	1.9 %	1.9 %
Median	6.9 %	3.5 %	3.3 %	3.9 %	3.8 %
Mean	8.7 %	5.5 %	5.4 %	4.9 %	4.9 %
3rd quartile	11.2 %	6.4 %	6.2 %	6.8 %	6.8 %
Maximum	84.4 %	69.1 %	66.3 %	30.8 %	30.4 %

Evidently, as the model complexity increases, its ability to accurately predict delivery tour length increases. Specifically, the mean APE reduces from 5.8 % for F_1C_0 to 4.1 % for F_1C_4. Similarly, the inter-quartile range on APE reduces from 5.4 % for F_1C_0 to 4.4 % for F_1C_4, with F_1C_3 having the smallest inter-quartile range of 4.2 %. Notably, both F_1C_3 and F_1C_4 introduce critical refinement as they incorporate normalized first moment of customer nearest neighbor distance – $\mu_s/\sqrt{a}$, to explain the variance in last-mile length parameter. In fact, only F_1C_4 model includes a higher moment, specifically, normalized second moment of the customer demand – σ_d/q_v, while F_1C_2 incorporates normalized first moment of the customer demand – μ_d/q_v. Overall, these results highlight the potential of these refinements in accurately predicting the length of the delivery tour in diverse logistic settings, however, as the model complexity increases, the risk of model overfitting the trained data increases as well. To this end, subsection 5.2 details a discussion on the validity of these models on benchmarked distribution environments.

F_2 Form: Much like the F_1 form models, F_2 form models (equations 19 – 23) also assume long-haul length to be proportional to unweighted average distance between the distribution facility and the customers – ρ_u. However, these models assume less-than-truckload travel with the delivery fleet performing $[n\mu_d/q_v]^+$ delivery tours to fulfil the expected total customer demand. Consequently, the long-haul length is given by $\rho_u[n\mu_d/q_v]^+$, traversed to-and-fro by the delivery fleet. Hereafter, symbolic regression develops five models with increasing complexity $(C_j; j \in [0,4])$, with model F_2C_0 akin to the Daganzo (1984) model. Here, Fig. 6 and Table 8 describe prediction accuracy of these five models through APE. As discussed earlier, increasing model complexity improves model accuracy in predicting delivery tour length. In particular, the mean APE reduces from 8.7 % for F_2C_0 to 4.9 % for F_2C_4. Likewise, the inter-quartile range on APE reduces from 8.3 % for F_2C_0 to 5.1 % for F_2C_4, with F_2C_3 having the smallest inter- quartile range of 5.0 %. Notably, in contrast to F_1 form models, the F_2 form models perform poorly in predicting the delivery tour length for the synthetic distribution environments. Specifically, the assumption of less-than-truckload travel overestimates the number of delivery tours, and consequently the total delivery tour length. Nonetheless, it is pertinent to test the models against benchmarked distribution environments for a comprehensive assessment. Again, refer to subsection 0 for a discussion on model validation.

F_3 Form: The F_3 form models (equations 24 – 28) assume long-haul length to be proportional to demand-weighted average distance between the distribution facility and the customers – ρ_w. Much like F_1 form models, these models consider full-truckload travel, where the delivery fleet performs $n\mu_d/q_v$ delivery tours to fulfill the expected total customer demand. Together, this renders a long-haul length of $\rho_w n\mu_d/q_v$, traversed to-and-fro by the delivery fleet. Subsequently, symbolic regression develops five models with increasing levels of complexity $(C_j; j \in [0,4])$, with model F_3C_0 akin to Daganzo (1984) model, as described earlier. Note, Fig. 7 and Table 9 describe prediction accuracy of these five models through APE distribution and associated statistics, respectively. As expected, the prediction accuracy of the model improves with increasing of model complexity. Specifically, the mean APE reduces from 5.8 % for F_3C_0 to 4.0 % for F_3C_4. Similarly, the inter-quartile range on APE reduces from 5.4 % for F_3C_0 to 4.4 % for F_3C_4, with F_1C_3 having the smallest inter-quartile range of 4.2 %. Notably, the predictive capability of F_3 form models is similar to that of F_1 form models. However, subsection 0 sheds further light on differences between these two model types through validation across benchmarked distribution environments.

F_4 Form: Like the F_3 form models, F_4 form models (equations 29 – 33) assume long-haul length to be proportional to demand-weighted average distance between the distribution facility and the customers – ρ_w. However, these models consider less-than-truckload travel with the delivery fleet performing $[n\mu_d/q_v]^+$ delivery tours to fulfil the expected total customer demand. Thus, the long-haul length is given by $\rho_u[n\mu_d/q_v]^+$, traversed to-and-fro by the delivery fleet. Hereafter, symbolic regression develops five models with increasing levels of complexity $(C_j; j \in [0,4])$, with model F_4C_0 akin to Daganzo (1984) model. As is the case with the other three model forms, both F_4C_3 and F_4C_4 incorporate normalized first moment of customer nearest neighbor distance – $\mu_s/\sqrt{a}$, to explain the variance in last-mile length parameter. Besides, F_4C_4 model includes normalized second moment of the customer demand – σ_d/q_v, while F_4C_2 incorporates normalized first moment of the customer demand – μ_d/q_v. Fig. 8 and Table 10 describe prediction accuracy of these five models through APE distribution and associated statistics, respectively. Again, as the model complexity increases, its ability to accurately predict delivery tour length for the synthetic distribution environment increases, as is the case with all other model forms. In particular, the mean APE reduces from 8.7 % for F_4C_0 to 4.9 % for F_4C_4. Likewise, the inter-quartile range on APE reduces from 8.4 % for F_4C_0 to 4.9 % for F_4C_3 and F_4C_4. Notably, the predictive

Table 11
Summary of Benchmark Datasets Used for Testing.

Dataset Source	Instances	Size
Arnold et al. (2019)	Antwerp1, Antwerp2, Brussels1, Brussels2, Flanders1, Flanders2, Ghent1, Ghent2, Leuven1, Leuven2	3000–10000
Augerat (1995)	P-n101-k4	100
Christofides and Eilon (1969)	E-n101-k8, E-n101-k14	100
Christofides et al. (1979)	CMT3, CMT4, CMT5, CMT8, CMT9, CMT10, CMT11, CMT12, CMT13, CMT14, M−n101−k10, M−n121−k7, M−n151−k12, M−n200−k16, M−n200−k17	100–199
DIMACS (2021)	Loggi-n401-k23, Loggi-n501-k24, Loggi-n601-k19, Loggi-n601-k42, Loggi-n901-k42, Loggi-n1001-k31, ORTEC-n242-k12, ORTEC-n323-k21, ORTEC-n405-k18, ORTEC-n455-k41, ORTEC-n510-k23, ORTEC-n701-k64	241–1000
Fisher (1994)	F-n135-k7	134
Golden et al. (1998)	Golden_1, Golden_2, Golden_3, Golden_4, Golden_5, Golden_6, Golden_7, Golden_8, Golden_9, Golden_10, Golden_11, Golden_12, Golden_13, Golden_14, Golden_15, Golden_16, Golden_17, Golden_18, Golden_19, Golden_20	200–483
Li et al. (2005)	Li_21, Li_22, Li_23, Li_24, Li_25, Li_26, Li_27, Li_28, Li_29, Li_30, Li_31, Li_32	560–1200
Rochat and Taillard (1995)	tai100a, tai100b, tai100c, tai100d, tai150a, tai150b, tai150c, tai150d, tai385	100–385
Uchoa et al. (2017)	X-n101-k25, X-n106-k14, X-n110-k13, X-n115-k10, X-n120-k6, X-n125-k30, X-n129-k18, X-n134-k13, X-n139-k10, X-n143-k7, X-n148-k46, X-n153-k22, X-n157-k13, X-n162-k11, X-n167-k10, X-n172-k51, X-n176-k26, X-n181-k23, X-n186-k15, X-n190-k8, X-n195-k51,X-n200-k36, X-n204-k19, X-n209-k16, X-n214-k11, X-n219-k73, X-n223-k34, X-n228-k23, X-n233-k16, X-n237-k14, X-n242-k48, X-n247-k50, X-n251-k28, X-n256-k16, X-n261-k13, X-n266-k58, X-n270-k35, X-n275-k28, X-n280-k17, X-n284-k15, X-n289-k60, X-n294-k50, X-n298-k31, X-n303-k21, X-n308-k13, X-n313-k71, X-n317-k53, X-n322-k28, X-n327-k20, X-n331-k15, X-n336-k84, X-n344-k43, X-n351-k40, X-n359-k29, X-n367-k17, X-n376-k94, X-n384-k52, X-n393-k38, X-n401-k29, X-n411-k19, X-n420-k130, X-n429-k61, X-n439-k37, X-n449-k29, X-n459-k26, X-n469-k138, X-n480-k70, X-n491-k59, X-n502-k39, X-n513-k21, X-n524-k153, X-n536-k96, X-n548-k50, X-n561-k42, X-n573-k30, X-n586-k159, X-n599-k92, X-n613-k62, X-n627-k43, X-n641-k35, X-n655-k131, X-n670-k130, X-n685-k75, X-n701-k44, X-n716-k35, X-n733-k159, X-n749-k98, X-n766-k71, X-n783-k48, X-n801-k40, X-n819-k171, X-n837-k142, X-n856-k95, X-n876-k59, X-n895-k37, X-n916-k207, X-n936-k151, X-n957-k87, X-n979-k58, X-n1001-k43	100–1000

capability of F_4 form models is similar to that of F_2 form models. However, like F_2 form models, F_4 form models tend to overestimate total delivery tour length, demonstrating that less-than-truckload assumption inflates prediction in synthetic environments. Nevertheless, it is important to test the models against benchmarked distribution environments for a comprehensive comparative assessment. To this end, the following subsection details the validity of these models to accurately predict the length of delivery tour for benchmarked distribution environments.

$$\mathbf{F_1C_0}: l = 2\rho_u \frac{n\mu_d}{q_v} + 0.572\frac{n}{\sqrt{\delta}} \tag{14}$$

$$\mathbf{F_1C_1}: l = 2\rho_u \frac{n\mu_d}{q_v} + 0.546\frac{n}{\sqrt{\delta}} \tag{15}$$

$$\mathbf{F_1C_2}: l = 2\rho_u \frac{n\mu_d}{q_v} + \left(\frac{\mu_d}{q_v} + 0.517\right)\frac{n}{\sqrt{\delta}} \tag{16}$$

$$\mathbf{F_1C_3}: l = 2\rho_u \frac{n\mu_d}{q_v} + \left(\frac{\mu_s}{\sqrt{a}} \Big/ \left(\frac{\mu_s}{\sqrt{a}} + 0.022\right)\right)\frac{n}{\sqrt{\delta}} \tag{17}$$

$$\mathbf{F_1C_4}: l = 2\rho_u \frac{n\mu_d}{q_v} + \left(\frac{\sigma_d}{q_v} + \left(\frac{\mu_s}{\sqrt{a}} \Big/ \left(\frac{\mu_s}{\sqrt{a}} + 0.023\right)\right)\right)\frac{n}{\sqrt{\delta}} \tag{18}$$

$$\mathbf{F_2C_0}: l = 2\rho_u \left[\frac{n\mu_d}{q_v}\right]^+ + 0.572\frac{n}{\sqrt{\delta}} \tag{19}$$

$$\mathbf{F_2C_1}: l = 2\rho_u \left[\frac{n\mu_d}{q_v}\right]^+ + 0.490\frac{n}{\sqrt{\delta}} \tag{20}$$

$$\mathbf{F_2C_2}: l = 2\rho_u \left[\frac{n\mu_d}{q_v}\right]^+ + \left(\frac{\mu_d}{q_v} + 0.462\right)\frac{n}{\sqrt{\delta}} \tag{21}$$

$$\mathbf{F_2C_3}: l = 2\rho_u \left[\frac{n\mu_d}{q_v}\right]^+ + \left(\frac{\mu_s}{\sqrt{a}} \Big/ \left(\frac{\mu_s}{\sqrt{a}} + 0.027\right)\right)\frac{n}{\sqrt{\delta}} \tag{22}$$

$$\mathbf{F_2C_4}: l = 2\rho_u \left[\frac{n\mu_d}{q_v}\right]^+ + \left(\frac{\sigma_d}{q_v} + \left(\frac{\mu_s}{\sqrt{a}} \Big/ \left(\frac{\mu_s}{\sqrt{a}} + 0.029\right)\right)\right)\frac{n}{\sqrt{\delta}} \tag{23}$$

$$\mathbf{F_3C_0}: l = 2\rho_w \frac{n\mu_d}{q_v} + 0.572\frac{n}{\sqrt{\delta}} \tag{24}$$

$$\mathbf{F_3C_1}: l = 2\rho_w \frac{n\mu_d}{q_v} + 0.546\frac{n}{\sqrt{\delta}} \tag{25}$$

$$\mathbf{F_3C_2}: l = 2\rho_w \frac{n\mu_d}{q_v} + \left(\frac{\mu_d}{q_v} + 0.517\right)\frac{n}{\sqrt{\delta}} \tag{26}$$

$$\mathbf{F_3C_3}: l = 2\rho_w \frac{n\mu_d}{q_v} + \left(\frac{\mu_s}{\sqrt{a}} \Big/ \left(\frac{\mu_s}{\sqrt{a}} + 0.022\right)\right)\frac{n}{\sqrt{\delta}} \tag{27}$$

$$\mathbf{F_3C_4}: l = 2\rho_w \frac{n\mu_d}{q_v} + \left(\frac{\sigma_d}{q_v} + \left(\frac{\mu_s}{\sqrt{a}} \Big/ \left(\frac{\mu_s}{\sqrt{a}} + 0.023\right)\right)\right)\frac{n}{\sqrt{\delta}} \tag{28}$$

$$\mathbf{F_4C_0}: l = 2\rho_w \left[\frac{n\mu_d}{q_v}\right]^+ + 0.572\frac{n}{\sqrt{\delta}} \tag{29}$$

$$\mathbf{F_4C_1}: l = 2\rho_w \left[\frac{n\mu_d}{q_v}\right]^+ + 0.490\frac{n}{\sqrt{\delta}} \tag{30}$$

$$\mathbf{F_4C_2}: l = 2\rho_w \left[\frac{n\mu_d}{q_v}\right]^+ + \left(\frac{\mu_d}{q_v} + 0.462\right)\frac{n}{\sqrt{\delta}} \tag{31}$$

$$\mathbf{F_4C_3}: l = 2\rho_w \left[\frac{n\mu_d}{q_v}\right]^+ + \left(\frac{\mu_s}{\sqrt{a}} \Big/ \left(\frac{\mu_s}{\sqrt{a}} + 0.027\right)\right)\frac{n}{\sqrt{\delta}} \tag{32}$$

$$\mathbf{F_4C_4}: l = 2\rho_w \left[\frac{n\mu_d}{q_v}\right]^+ + \left(\frac{\sigma_d}{q_v} + \left(\frac{\mu_s}{\sqrt{a}} \Big/ \left(\frac{\mu_s}{\sqrt{a}} + 0.029\right)\right)\right)\frac{n}{\sqrt{\delta}} \tag{33}$$

5.2. Testing Set

Having trained the continuous approximation models on 1080 synthetic distribution environments, this study tests the effectiveness of these models in estimating delivery tour length for the following 182 benchmarked distribution environments (Table 11).

Consequently, Figs. 9 – 12 and Tables 12 – 15 showcase the absolute percentage error boxplot distributions and corresponding statistical measures, respectively, for F_1, F_2, F_3 and F_4 form models, each, against

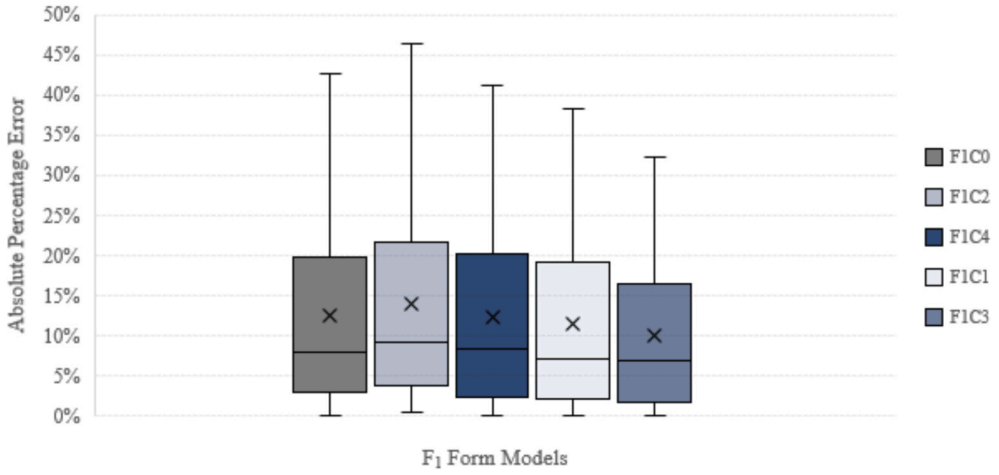


Fig. 9. Box Plot – Absolute Percentage Error of F_1 Form Models on Testing Set.

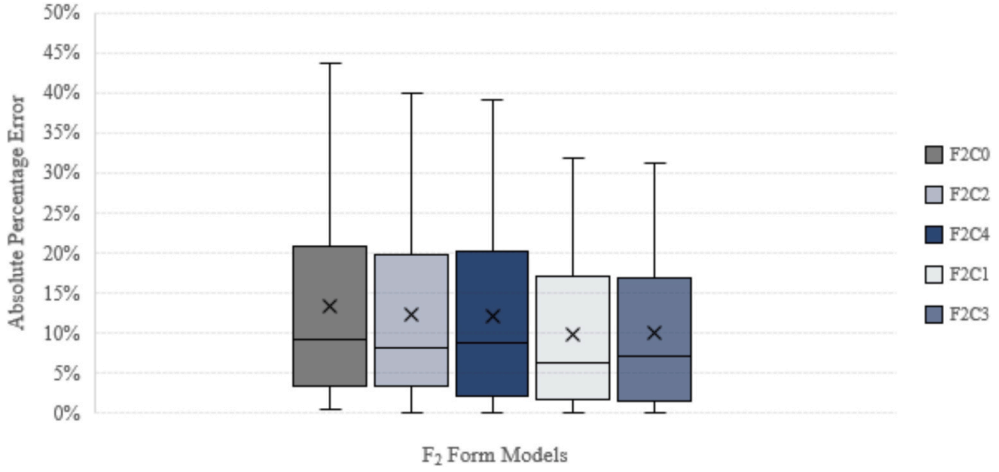


Fig. 10. Box Plot – Absolute Percentage Error of F_2 Form Models on Testing Set.

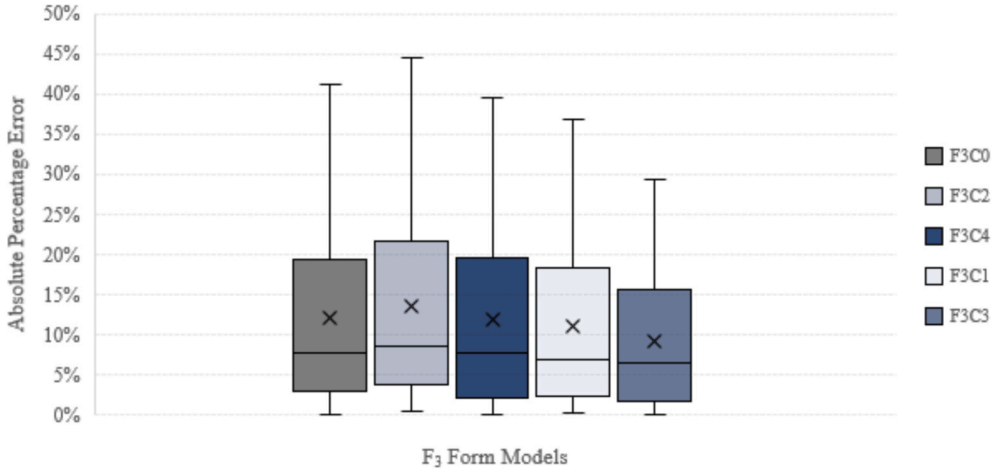


Fig. 11. Box Plot – Absolute Percentage Error of F_3 Form Models on Testing Set.

this testing set. Note, for the purpose of visualization, these figures and tables present the outcomes in the order of decreasing maximum absolute percentage error, rendering the following order (right to left): F_iC_3, F_iC_1, F_iC_4, and F_iC_2, contrasted against F_iC_0.

Notably, across different model forms the performance disparity against benchmarked distribution environments is less pronounced than that observed in the previous subsection with synthetic distribution environments. Further, in the context of model complexity, F_iC_1 and F_iC_3 models outperformed other models in accurately predicting the length of the delivery tour, across all model forms. These results confirm the

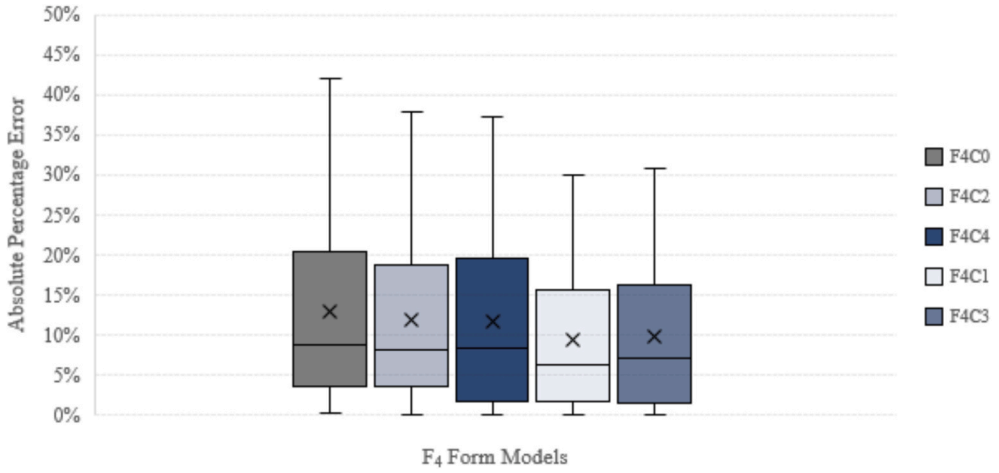


Fig. 12. Box Plot – Absolute Percentage Error of F_4 Form Models on Testing Set.

Table 12
Descriptive Statistics – Absolute Percentage Error of F_1 Form Models on Testing Set.

Measure	F_1C_0	F_1C_2	F_1C_4	F_1C_1	F_1C_3
Minimum	0.1 %	0.4 %	0.0 %	0.0 %	0.1 %
1st quartile	3.8 %	5.0 %	3.0 %	2.9 %	2.3 %
Median	6.4 %	8.0 %	6.9 %	5.9 %	5.9 %
Mean	9.2 %	10.3 %	9.7 %	8.4 %	8.0 %
3rd quartile	12.3 %	13.5 %	13.2 %	12.8 %	11.3 %
Maximum	42.7 %	46.4 %	41.2 %	38.4 %	32.2 %

Table 13
Descriptive Statistics – Absolute Percentage Error of F_2 Form Models on Testing Set.

Measure	F_2C_0	F_2C_2	F_2C_4	F_2C_1	F_2C_3
Minimum	0.4 %	0.0 %	0.1 %	0.1 %	0.1 %
1st quartile	4.4 %	4.4 %	2.7 %	2.4 %	2.1 %
Median	8.0 %	6.9 %	7.8 %	4.8 %	6.0 %
Mean	10.3 %	9.5 %	9.8 %	7.7 %	8.2 %
3rd quartile	13.2 %	13.0 %	13.8 %	12.3 %	12.0 %
Maximum	43.6 %	39.9 %	39.1 %	31.9 %	31.3 %

Table 14
Descriptive Statistics – Absolute Percentage Error of F_3 Form Models on Testing Set.

Measure	F_3C_0	F_3C_2	F_3C_4	F_3C_1	F_3C_3
Minimum	0.1 %	0.4 %	0.1 %	0.2 %	0.1 %
1st quartile	3.8 %	5.0 %	2.9 %	3.0 %	2.2 %
Median	6.4 %	7.2 %	6.1 %	5.6 %	5.4 %
Mean	9.0 %	10.1 %	9.3 %	8.2 %	7.6 %
3rd quartile	12.2 %	14.0 %	13.0 %	12.2 %	11.2 %
Maximum	41.2 %	44.5 %	39.5 %	36.9 %	29.3 %

Table 15
Descriptive Statistics – Absolute Percentage Error of F_4 Form Models on Testing Set.

Measure	F_4C_0	F_4C_2	F_4C_4	F_4C_1	F_4C_3
Minimum	0.2 %	0.1 %	0.1 %	0.0 %	0.0 %
1st quartile	4.7 %	4.7 %	2.3 %	2.3 %	2.0 %
Median	7.6 %	6.8 %	7.3 %	5.1 %	6.3 %
Mean	10.0 %	9.3 %	9.3 %	7.6 %	7.7 %
3rd quartile	13.1 %	12.3 %	13.6 %	11.0 %	11.5 %
Maximum	42.1 %	37.9 %	37.2 %	29.9 %	30.8 %

Table 16
Descriptive Statistics – Absolute Percentage Error of Pareto-Optimal Models on Testing Set.

Measure	F_1C_0	F_1C_3	F_2C_1	F_3C_3	F_4C_1
Minimum	0.1 %	0.1 %	0.1 %	0.0 %	0.1 %
1st quartile	2.3 %	2.4 %	2.2 %	2.3 %	2.3 %
Median	5.9 %	4.8 %	5.4 %	5.1 %	5.9 %
Mean	8.0 %	7.7 %	7.6 %	7.6 %	8.0 %
3rd quartile	11.3 %	12.3 %	11.2 %	11.0 %	11.3 %
Maximum	32.2 %	31.9 %	29.3 %	29.9 %	32.2 %

underfitting tendencies of F_iC_2 models, which lack sufficient sophistication to capture the variance in distribution length. Additionally, the results affirm the overfitting nature of F_iC_4 models, as they are too closely tailored to synthetic training data and thus falter in generalization. In fact, F_2C_1 and F_4C_1 have a mean absolute percentage error of 7.7 % and 7.6 %, respectively, least among other models of the same form. Similarly, F_1C_3 and F_3C_3 have the smallest mean absolute percentage errors among other models of the same form – 8.0 % and 7.6, respectively. Together, these four models form the pareto-optimal models (refer to Table 16 and Fig. 13), considering that they are statistically significantly different from the baseline model – F_1C_0, formulated by Daganzo (1984), but statistically indistinct from each other (Table 17). However, in the context of model application F_2C_1 and F_3C_3 prove to be more useful than the other pareto-optimal models. Specifically, when the logistic operator has a specific fleet of $m = [n\mu_d/q_v]^+$ delivery vehicles to operate but no information on customers' spatial dispersion and demand pattern, model $F_2C_1 : l = 2\rho_u[n\mu_d/q_v]^+ + 0.490n/\sqrt{\delta}$ is recommended. On the other hand, if the logistic operator has complete information on customers' spatial and demand characteristics, this study recommends use of model $F_3C_3 : l = 2\rho_w(n\mu_d/q_v) + ((\mu_s/\sqrt{a})/(\mu_s/\sqrt{a} + 0.022))\left(n/\sqrt{\delta}\right)$ to estimate delivery tour length. Together, these models make substantial improvements on continuous approximation-based estimation of delivery tour length, enabling strategic decision-making/planning for the logistic operator.

6. Conclusions

The conventional approach to modelling last-mile distribution operations typically relies on discrete formulations, which, while mathematically rigorous, require sophisticated solution techniques. This approach is warranted in operational planning contexts where decision-makers have exact information and seek precise plans. However, in strategic planning contexts, where available information is often representative but not necessarily exact, such level of precision becomes

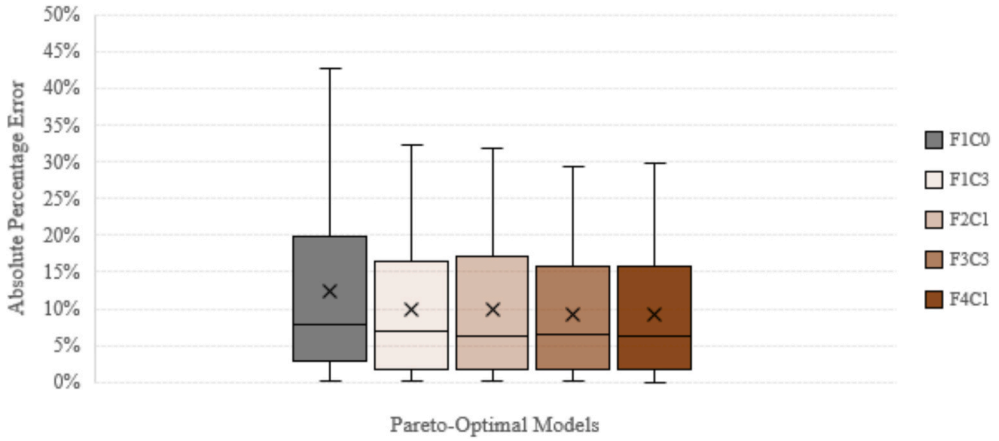


Fig. 13. Box Plot – Absolute Percentage Error of Pareto-Optimal Models on Testing Set.

Table 17
Test for Statistical Differences among Pareto-Optimal Models.

p-values	F_1C_0	F_1C_3	F_2C_1	F_3C_3	F_4C_1
F_1C_0	–	0.016	0.000	0.001	0.000
F_1C_3	0.016	–	0.468	0.106	0.496
F_2C_1	0.000	0.468	–	0.787	0.871
F_3C_3	0.001	0.106	0.787	–	0.860
F_4C_1	0.000	0.496	0.871	0.860	–

redundant. To this end, continuous approximation (CA) method offers a practical alternative. By leveraging continuous density functions, CA models offer tractable and interpretable formulations that estimate parameters approximately, thus balancing estimation accuracy and computational effort. Nevertheless, existing CA-based routing frameworks assume an overly simplified distribution environment with a distribution facility located sufficiently far from the service region, catering to customers having uniformly dispersed random locations, and fixed unitary demand. These assumptions significantly limits the capability of CA frameworks to accurately model last-mile operations in real-world distribution environments, such as conventional e-commerce (apparels, accessories, electronics, etc.), quick-commerce (groceries, meals, medicines, daily essentials, etc.), bulk-commerce (furniture, home appliances, etc.), as well as postal service, waste collection, and emergency service, among others. In fact, these real-world distribution environments may have peculiar distribution structure and distinctive customer characteristics that significantly influence delivery patterns and vehicle routing strategies. Thus, to address the limitations of extant CA models, this work developed a holistic CA framework for Vehicle Routing Problem (VRP) to facilitate strategic last-mile distribution planning within diverse logistic settings, each with specific facility location, unique spatial dispersion, and distinct demand pattern. Specifically, this study synthesized a total of 1080 such unique distribution environments, subsequently, optimized last-mile distribution for each distribution environment instance using an Adaptive Large Neighborhood Search (ALNS) metaheuristic, and finally developed a set of robust CA-based functional forms through symbolic regression to precisely estimate the total distribution tour length across these synthesized distribution environments.

Model 1: low-fidelity model under minimal information

When the logistic operator has aggregate system level information, including expected number of customers – n, customer density – δ, distance from the facility to the service region – ρ, and fleet size – m, but lacks information on customers' spatial dispersion and demand pattern, the following model proves to be effective in predicting the delivery tour length:

$$l = 2\rho m + 0.49n\Big/\sqrt{\delta} \tag{34}$$

Notably, this model is structurally equivalent to the baseline Daganzo (1984) model, albeit achieving a statistically smaller error (mean APE) of 7.7 % in contrast to 9.2 % for the baseline model. Thus, despite its simplicity, this model demonstrates strong predictive performance and can guide the logistics operator to optimally design service area, locate facility, and deploy delivery fleet at early planning stages, when granular data is unavailable. It is particularly valuable in greenfield expansion projects with logistic operator entering a new market, having no operational history. Further, national postal services planning expansions into rural regions may only know approximate parcel volumes, rendering this low-fidelity model suitable for application. The model is also relevant to emergency service logistics (e.g., ambulance deployment), where high-resolution demand and location information is typically unavailable, thereby facilitating a tractable basis for early-stage decisions on fleet sizing and service region design.

Model 2: high-fidelity model under rich information

On the other hand, if the logistic operator has complete information on customers' spatial and demand characteristics, this study recommends the following model to estimate delivery tour length:

$$l = 2\rho_w(n\mu_d/q_v) + \left((\mu_s/\sqrt{a})/(\mu_s/\sqrt{a} + 0.022)\right)\left(n\Big/\sqrt{\delta}\right) \tag{35}$$

This high-fidelity model enables more precise cost forecasting, resource allocation, and scenario evaluation where market analytics is available. Specifically, the model renders a smaller mean APE of 7.6 % with a narrower APE range of 29.2 %, compared to 8.2 % and 41.1 %, respectively, for the baseline Daganzo (1984) model. This advancement establishes improved predictive accuracy of CA models in estimating distribution tour length, thus enhancing their applicability in strategic decision-making. Consequently, industries with access to rich data can leverage this high-fidelity model for more precise evaluations. For instance, e-commerce firms that maintain detailed information on customer demand (location, frequency, and lead-time, among others) can utilize this model to refine fleet size, micro-consolidation center and collection-point locations, as well as service area partitioning. Likewise, urban waste management systems with geocoded household and waste generation data can use this model to optimize routing efficiency and accurately forecast operational costs.

Overall, this work extends the scope and applicability of CA models, enabling decision-makers to approximate logistics operations, and consequently, distribution costs across heterogeneous environments, even in the absence of granular operational data. In doing so, the study offers an interpretable, tractable, and computationally efficient (holistic) CA framework for strategic last-mile distribution planning. Future

work could focus on expanding this framework to include more complex routing constraints including multi-echelon distribution, delivery within specified time-windows, simultaneous pickup and delivery operations, and heterogenous delivery fleets, among others. These enhancements may be achieved through incremental analytical development of the framework, thereby expanding its scope, or alternatively, through an independent exploration of a closed-form framework via symbolic regression, consistent with the methodology adopted in this work. Nevertheless, while the symbolic-regression-based CA models provide analytical transparency, they may encounter limitations in capturing more complex, nonlinear interactions, especially as systems become more sophisticated. This opens promising avenues for future research involving data-driven, nonparametric frameworks, particularly neural networks. Although inherently non-interpretable, such models have the potential to capture higher-order dependencies and context-specific patterns with far greater predictive precision than symbolic regression. To this end, future studies could benchmark parametric and nonparametric methods to underscore the trade-offs between the two, allowing logistic operators to choose between interpretability and accuracy depending on the context of decision-making – strategic, tactical, or operational. Perhaps, a hybrid modelling paradigm that blends the mathematical interpretability of parametric models with the learning capacity of nonparametric models could represent the next frontier in strategic decision-making associated with operations research (Cranmer et al., 2020). Nonetheless, metaheuristics will remain central to operational decision-making. Among them, ALNS has demonstrated success in solving wide range of routing problems, however, it is not without its limitations. Specifically, its large number of parameters introduces significant degrees of freedom, making calibration and validation challenging. Further, it lacks a guiding framework for design, development, and deployment of operators for different problem contexts, making systematic implementation difficult. Addressing these challenges presents an important direction for future work in operations research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

None

References

Anand, N., Van Duin, R., Quak, H., Tavasszy, L., 2015. Relevance of city logistics modelling efforts: a review. Transp. Rev. 35 (6), 701–719.

Ansari, S., Başdere, M., Li, X., Ouyang, Y., Smilowitz, K., 2018. Advancements in continuous approximation models for logistics and transportation systems: 1996–2016. Transp. Res. B Methodol. 107, 229–252.

Arnold, F., Gendreau, M., Sörensen, K., 2019. Efficiently solving very large-scale routing problems. Comput. Oper. Res. 107, 32–42.

Augerat, P., 1995. Approche polyèdrale du problème de tournées de véhicules. Institut National Polytechnique De. Grenoble-INPG.

Azi, N., Gendreau, M., Potvin, J.-Y., 2014. An adaptive large neighborhood search for a vehicle routing problem with multiple routes. Comput. Oper. Res. 41, 167–173.

Beardwood, J., Halton, J.H., Hammersley, J.M., 1959. The shortest path through many points, *Mathematical proceedings of the Cambridge philosophical society*. Cambridge University Press 299–327.

Bezanson, J., Edelman, A., Karpinski, S., Shah, V.B., 2017. Julia: a fresh approach to numerical computing. SIAM Rev. 59 (1), 65–98.

Braekers, K., Ramaekers, K., Van Nieuwenhuyse, I., 2016. The vehicle routing problem: State of the art classification and review. Comput. Ind. Eng. 99, 300–313.

Capgemini, 2019. Share of total supply costs worldwide in 2018, by type of cost.

Çavdar, B., Sokol, J., 2015. A distribution-free TSP tour length estimation model for random graphs. Eur. J. Oper. Res. 243 (2), 588–598.

Christofides, N., Eilon, S., 1969. An algorithm for the vehicle-dispatching problem. J. Oper. Res. Soc. 20 (3), 309–318.

Christofides, N., Mingozzi, A., Toth, P., 1979. The vehicle routing problem, In: Christofides, N., Mingozzi, A., Toth, P., Sandi, C. (Eds.), Combinatorial optimization. Wiley Interscience, pp. 315–338.

Contardo, C., Hemmelmayr, V., Crainic, T.G., 2012. Lower and upper bounds for the two-echelon capacitated location-routing problem. Comput. Oper. Res. 39 (12), 3185–3199.

Cranmer, M., Sanchez Gonzalez, A., Battaglia, P., Xu, R., Cranmer, K., Spergel, D., Ho, S., 2020. Discovering symbolic models from deep learning with inductive biases. Adv. Neural Inf. Proces. Syst. 33, 17429–17442.

Daganzo, C.F., 1984. The distance traveled to visit N Points with a maximum of C stops per vehicle: an analytic model and an application. Transp. Sci. 18 (4), 331–350.

Davis, B.A., Figliozzi, M.A., 2013. A methodology to evaluate the competitiveness of electric delivery trucks. Transp. Res. Part e: Logistics Transp. Rev. 49 (1), 8–23.

DIMACS, 2021. Implementation Challenge: Vehicle Routing.

Elshaer, R., Awad, H., 2020. A taxonomic review of metaheuristic algorithms for solving the vehicle routing problem and its variants. Comput. Ind. Eng. 140, 106242.

Estrada, M., Roca-Riu, M., 2017. Stakeholder's profitability of carrier-led consolidation strategies in urban goods distribution. Transp. Res. Part e: Logistics Transp. Rev. 104, 165–188.

Figliozzi, M.A., 2008. Planning approximations to the average length of vehicle routing problems with varying customer demands and routing constraints. Transp. Res. Record: J. Transp. Res. Board 2089 (1), 1–8.

Figliozzi, M.A., 2009. Planning approximations to the average length of vehicle routing problems with time window constraints. Transp. Res. B Methodol. 43 (4), 438–447.

Figliozzi, M.A., 2010. The impacts of congestion on commercial vehicle tour characteristics and costs. Transportation Research Part e: Logistics and Transportation Review 46 (4), 496–506.

Fisher, M.L., 1994. Optimal solution of vehicle routing problems using minimum k-trees. Oper. Res. 42 (4), 626–642.

Franceschetti, A., Jabali, O., Laporte, G., 2017. Continuous approximation models in freight distribution management. TOP 25 (3), 413–433.

Ghaffarinasab, N., Van Woensel, T., Minner, S., 2018. A continuous approximation approach to the planar hub location-routing problem: Modeling and solution algorithms. Comput. Oper. Res. 100, 140–154.

Golden, B.L., Wasil, E.A., Kelly, J.P., Chao, I.-M., 1998. The impact of metaheuristics on solving the vehicle routing problem: algorithms, problem sets, and computational results. Fleet Management and Logistics. Springer 33–56.

Grangier, P., Gendreau, M., Lehuédé, F., Rousseau, L.-M., 2016. An adaptive large neighborhood search for the two-echelon multiple-trip vehicle routing problem with satellite synchronization. Eur. J. Oper. Res. 254 (1), 80–91.

Konstantakopoulos, G.D., Gayialis, S.P., Kechagias, E.P., 2020. Vehicle routing problem and related algorithms for logistics distribution: a literature review and classification. Oper. Res. 1–30.

Langevin, A., Mbaraga, P., Campbell, J.F., 1996. Continuous approximation models in freight distribution: an overview. Transp. Res. B Methodol. 30 (3), 163–188.

Li, F., Golden, B., Wasil, E., 2005. Very large-scale vehicle routing: new test problems, algorithms, and results. Comput. Oper. Res. 32 (5), 1165–1179.

Mardešić, N., Erdelić, T., Carić, T., Đurasević, M., 2023. Review of Stochastic Dynamic Vehicle Routing in the Evolving Urban Logistics Environment. Mathematics 12 (1), 28.

Merchán, D., Winkenbach, M., 2019. An empirical validation and data-driven extension of continuum approximation approaches for urban route distances. Networks 73 (4), 418–433.

Nicola, D., Vetschera, R., Dragomir, A., 2019. Total distance approximations for routing solutions. Comput. Oper. Res. 102, 67–74.

Ouyang, Y., Daganzo, C.F., 2006. Discretization and Validation of the Continuum Approximation Scheme for Terminal System Design. Transp. Sci. 40 (1), 89–98.

Pahwa, A., 2023. Assessing Sustainability of E-Commerce Goods distribution. UC Davis.

Pahwa, A., Jaller, M., 2022. A cost-based comparative analysis of different last-mile strategies for e-commerce delivery. Transportation Research Part e: Logistics and Transportation Review 164, 102783.

Prodhon, C., Prins, C., 2014. A survey of recent research on location-routing problems. Eur. J. Oper. Res. 238 (1), 1–17.

Rifki, O., Garaix, T., 2023. Traveling salesman problem with time slots: Asymptotic analysis and resolution algorithm. *arXiv preprint arXiv:2303.13998*.

Rifki, O., Garaix, T., Solnon, C., 2021. An asymptotic approximation of the traveling salesman problem with uniform non-overlapping time windows. In: *2021 IEEE 17th International Conference on Automation Science and Engineering (CASE)*. IEEE, pp. 983–988.

Rios, B.H.O., Xavier, E.C., Miyazawa, F.K., Amorim, P., Curcio, E., Santos, M.J., 2021. Recent dynamic vehicle routing problems: a survey. Comput. Ind. Eng. 160, 107604.

Rochat, Y., Taillard, É.D., 1995. Probabilistic diversification and intensification in local search for vehicle routing. J. Heuristics 1, 147–167.

Ropke, S., Pisinger, D., 2006. An adaptive large neighborhood search heuristic for the pickup and delivery problem with time windows. Transp. Sci. 40 (4), 455–472.

Saberi, M., Verbas, İ.Ö., 2012. Continuous Approximation Model for the Vehicle Routing Problem for Emissions Minimization at the Strategic Level. J. Transp. Eng. 138 (11), 1368–1376.

Schneider, M., Drexl, M., 2017. A survey of the standard location-routing problem. Ann. Oper. Res. 259 (1), 389–414.

Turkensteen, M., Klose, A., 2012. Demand dispersion and logistics costs in one-to-many distribution systems. Eur. J. Oper. Res. 223 (2), 499–507.

Uchoa, E., Pecin, D., Pessoa, A., Poggi, M., Vidal, T., Subramanian, A., 2017. New benchmark instances for the capacitated vehicle routing problem. Eur. J. Oper. Res. 257 (3), 845–858.

Universal Postal Union, 2021. Number of parcels distributed worldwide from 2015 to 2020, by sector (in million units).

Zhang, H., Ge, H., Yang, J., Tong, Y., 2021. Review of Vehicle Routing Problems: Models, Classification and solving Algorithms. Arch. Comput. Meth. Eng. 1–27.